

Large-Scale Irrigation and Water Use Inequality in South Africa

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Abstract

Large-scale irrigation infrastructure can raise agricultural productivity, yet only 3.5% of agricultural land in Africa is irrigated. This paper evaluates the distributional impacts of large-scale surface irrigation in South Africa. Using quasi-experimental variation in elevation around canals, I estimate effects of irrigation infrastructure separately for subsistence farmers and for commercial farmers. Commercial farmers increase their crop yields and expand cultivated land. Using census data, I find that the expanding commercial farms generate rural employment. In contrast, subsistence farmers experience declines in crop yields and do not expand cultivated area. Suggestive evidence highlights the role of capital constraints and reduced river water availability. Thus, infrastructure alone is insufficient to raise agricultural productivity among subsistence farmers. Finally, a cost-benefit analysis shows that the aggregate gains are large, even though benefits accrue disproportionately to commercial farmers. The coexistence of aggregate gains and adverse distributional impacts raises a fundamental question of efficiency-equity tradeoffs in large public infrastructure investments.

Keywords: Irrigation, Public Infrastructure, Regression Discontinuity, Agricultural Productivity, Land Use, Remote Sensing, South Africa

JEL codes: O13, Q15, Q24, Q25

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1 Introduction

Agricultural productivity growth is central to economic development. A large body of theoretical work argues that productivity gains in agriculture are a necessary precondition for structural transformation and long-run growth because they release labor and capital toward higher-productivity sectors (Lewis, 1954; Johnston and Mellor, 1961; Schultz, 1964; Matsuyama, 1992; Gollin, Parente and Rogerson, 2002). Empirically, differences in agricultural productivity account for a substantial share of cross-country income gaps (Restuccia, Yang and Zhu, 2008). The problem is especially acute in Sub-Saharan Africa, where agricultural productivity is the lowest in the world (Gollin, Lagakos and Waugh, 2014; Suri and Udry, 2022). Yet one of the most productive agricultural technologies — irrigation — remains vastly underutilized on the continent. Only 3.5 percent of Africa’s agricultural land is irrigated (Foster and Briceño-Garmendia, 2010), leaving substantial scope for expansion. Realizing this potential requires large-scale public infrastructure investment, most notably the construction of irrigation dams that can store water in periods of abundant rainfall for later use in times of drought. Such investments have historically been controversial due to adverse distributional impacts on upstream areas (WCD, 2000; Duflo and Pande, 2007). Nevertheless, international donors have recently renewed their commitment to financing large dam projects (Foster and Briceño-Garmendia, 2010; Dillon and Fishman, 2019). Whether large-scale irrigation infrastructure can deliver the intended improvements in agricultural productivity in Sub-Saharan Africa remains an open question.

This paper contributes to the ongoing debate on whether large-scale surface irrigation constitutes a worthwhile public investment for economic development, with particular relevance for Sub-Saharan Africa. A central and underexplored question is whether the benefits of irrigation accrue to all farmers who gain access to it, or whether resource constraints prevent some from benefiting. The agricultural sector in sub-Saharan Africa is characterized by the coexistence of resource-poor subsistence farmers and more productive commercial operations. Previous literature has documented a wide range of constraints that prevent resource-poor farmers from taking advantage of productive investments, including credit and capital constraints (Karlan et al., 2014; Beaman et al., 2023), input access constraints (Suri, 2011; Duflo, Kremer and Robinson, 2011; Bold et al., 2017), information and learning frictions (Foster and Rosenzweig, 1995; Conley and Udry, 2010; BenYishay and Mobarak, 2019; Beaman et al., 2021), insecure land tenure (Goldstein and Udry, 2008), limited market access (Bergquist and Dinerstein, 2020), and labor market failures (Jones et al., 2022). These same constraints may prevent subsistence farmers

from benefiting from large-scale irrigation infrastructure.

South Africa offers a unique setting to study this question. Its agricultural sector has a dual structure in which resource-poor subsistence farmers, comparable to smallholders elsewhere in Sub-Saharan Africa, coexist with highly productive, export-oriented commercial farms. Understanding whether these two types of farmers benefit differentially from access to large-scale irrigation infrastructure speaks directly to whether this type of investment can be scaled up and expected to increase agricultural productivity and eventually start the structural transformation process across sub-Saharan Africa.

My identification strategy exploits the gravity-driven nature of surface irrigation within a regression discontinuity (RD) framework (Asher et al., 2024). Water in canals flows downhill, so fields located below a canal can be irrigated by gravity at a low cost, while fields above a canal require energy-intensive upward pumping. This generates a sharp discontinuity in the cost of irrigation at the threshold of zero relative elevation to the nearest canal, which in turn produces a corresponding discontinuity in the probability of land being irrigated. I exploit this discontinuity using an RD design, with relative elevation to the nearest canal as the running variable, considering the areas below a canal as treated and the areas above as the counterfactual.

The RD design requires units of analysis at a resolution fine enough to make treatment assignment unambiguous. Therefore, I rely on 30×30 m grid cells derived from satellite imagery which make it more likely each unit lies entirely above or entirely below the threshold. This represents an improvement over coarser units of analysis, such as villages or districts, which may span both sides of the threshold. I merge data from multiple sources to construct a dataset of over 29 million observations, including information on crop productivity and land use. Combining these data with the coordinates and elevation of the starting points of 144 irrigation canals allows me to assign treatment status at the field or even sub-field level. When analyzing crop yields, I restrict the sample to grid cells classified as growing annual crops, thereby removing noise from non-crop vegetation and perennial crops. For socioeconomic outcomes, I merge census data at the small area level with an elevation raster and canal coordinates, yielding a sample of 3,035 observations.¹ Both datasets are overlaid with the boundaries of former homelands, thus allowing for the estimation of heterogeneous treatment effects by farmer type, since subsistence farmers are almost exclusively concentrated in the former homelands.²

¹The South Africa 2011 Census at the small area level covers 100 percent of the population and contains 84,907 census units. The small area level falls between the enumeration area and sub-place levels.

²Combining the homeland boundaries with a 73-class land cover classification dataset that distinguishes between subsistence farmers and commercial farmers, I find that subsistence farmers are concentrated almost exclusively in the former homelands, while commercial farms are located in non-homeland areas.

I find that irrigation canals raise agricultural production both at the intensive margin (through higher crop yields) and the extensive margin (through expansion of cultivated area). However, these gains accrue entirely to commercial farmers in non-homeland areas. Below canals in non-homelands, wheat yields increase by 3.9 percent and maize yields by 2.3 percent. In former homelands, in contrast, wheat yields fall by 2.9 percent and maize yields by 2.6 percent, with the maize decline statistically significant at conventional levels. At the extensive margin, downstream areas in non-homelands are 4.26 percentage points more likely to be agricultural (a 20 percent increase relative to the control mean) driven primarily by an expansion in annual crops (up by 4.53 pp, or 27 percent) and irrigated annual crops in particular (up 2.27 pp, or 54 percent). In former homelands, no such expansion occurs. The evidence thus indicates that subsistence farmers fail to benefit from large-scale irrigation technology to the same extent as their commercial counterparts. These results are robust to extending the analysis to a longer time horizon, varying the vertical and horizontal distance bandwidths, excluding a donut hole around the threshold, and controlling for soil quality.

I provide suggestive evidence on two mechanisms behind the absence of productivity gains for subsistence farmers. First, capital constraints result in the absence of last-mile infrastructure connecting farmers to the main canal. If the costs of constructing and maintaining secondary distribution networks increase with distance from the canal head, the benefits of canal exposure should decline with that distance in homeland areas but not in non-homelands where capital constraints do not bind. Consistent with this prediction, I find that treatment effects in former homelands decrease monotonically with distance to the canal head and eventually turn negative, while in non-homelands treatment effects are stable across the same distance gradient. The second mechanism is water diversion from rivers. When canal intakes divert river water, farmers downstream from canal heads who rely on rivers as their primary irrigation source face reduced water availability. Under this mechanism, treatment effects should worsen with proximity to rivers in homelands but not in non-homelands. I find precisely this pattern: subsistence farmers located near rivers experience significantly larger yield losses than those further away. I complement this evidence with data from the South African water user database, showing that farmers in former homelands are substantially more likely to rely on river water for irrigation. I further confirm the physical mechanism using monthly discharge data from river gauging stations: river segments downstream of canal heads carry significantly less water during peak irrigation months relative to upstream segments on the same river, while no such differential is observed outside the irrigation season.

To assess whether agricultural productivity gains translate into broader welfare im-

provements, I estimate the impact of irrigation canals on local labor markets using the same RD design applied to census data. Downstream areas in non-homelands exhibit a 4.9 percentage point lower unemployment rate (a 20 percent reduction relative to the control group) and a 2.7 percentage point decline in the share of households with zero income. In former homelands, treatment effects on labor market outcomes are small and statistically insignificant.

Finally, I combine the RD estimates with cost projections from a government feasibility study to conduct a cost-benefit analysis of a proposed large irrigation dam in Limpopo province. In the baseline scenario with a 3 percent discount rate, I obtain a benefit-cost ratio of 1.62, suggesting that large irrigation dams constitute a cost-effective investment even after accounting for the negative effects imposed on subsistence farmers. The ratio remains above one under a 4 percent discount rate. However, in a scenario where the dam benefits only existing agricultural land, and does not allow for conversion of uncultivated land, the ratio falls below one, indicating that the investment would not recover its costs. Two policy implications follow. First, these findings highlight a sharp efficiency-equity tradeoff in large-scale public infrastructure provision, which requires effective redistribution policy. Subsistence farmers should be compensated for the negative externalities they bear. Second, the economic viability of new dam construction depends critically on whether the command area encompasses convertible uncultivated land; investments that do not lead to the agricultural land expansion may not be justified.

This paper makes three contributions to the existing literature. First, I add to the literature on the long-run impacts of large-scale irrigation infrastructure (Duflo and Pande, 2007; Zaveri, Russ and Damania, 2020; Dillon and Fishman, 2019; Hornbeck and Keskin, 2014; Strobl and Strobl, 2011; Olmstead and Sigman, 2015; Blanc and Strobl, 2014; Metetal, 2019). Much of this literature relies on instrumental variable strategies that exploit geographic suitability for dam construction. More recently, regression discontinuity designs using distance to command area boundaries (Blakeslee et al., 2023; Jones et al., 2022; Hagerty, 2022) or relative elevation to the nearest canal (Asher et al., 2024) have provided more credible local estimates by relying on weaker identifying assumptions. I contribute to this growing literature by documenting distributional impacts across farmer types,³ applying the Asher et al. (2024) RD design in a new context, examining land use outcomes from a novel fine-grained classification dataset, and conducting a comprehensive cost-benefit analysis that incorporates distributional effects. This last element has

³The distributional impacts in Duflo and Pande (2007) refer to differential effects across districts downstream of dams relative to districts hosting the dam, rather than across farmer types within the command area.

been identified as a critical gap in prior work (Dillon and Fishman, 2019). Second, I contribute to the literature on the role of public infrastructure in economic development. Large-scale electrification has been shown to generate positive welfare effects (Dinkelman, 2011; Lipscomb, Mobarak and Barham, 2013). Transportation infrastructure generally improves welfare in both developed and developing countries (Donaldson and Hornbeck, 2016; Donaldson, 2018; Burgess and Donaldson, 2010; Morten and Oliveira, 2024; Ghani, Goswami and Kerr, 2016; Baum-Snow et al., 2017; Storeygard, 2016; Banerjee, Duflo and Qian, 2012; Jedwab and Storeygard, 2022), though a subset of studies finds null effects in rural and remote areas (Asher and Novosad, 2020; Faber, 2014). I add to this body of work by documenting that large irrigation infrastructure investments fail to generate welfare improvements in a historically disadvantaged region populated by resource-poor farmers. Third, I contribute to the literature on the persistence of place-based policies and path dependence of distribution of economic activity (Bleakley and Lin, 2012; Jedwab and Moradi, 2016). Former homelands continue to lag behind the rest of South Africa in labor market outcomes (Lochmann, 2022), and individuals relocated to these regions in childhood exhibit lower educational attainment (Carrillo, Charris and Iglesias, 2023). This paper adds to the evidence on the enduring legacy of apartheid by demonstrating that farmers in former homelands remain unable to benefit from large-scale irrigation investments.

The paper is organized as follows. In Section 2, I provide background information on irrigation infrastructure and water policy in South Africa. In Section 3, I describe the data. Section 4 presents the empirical strategy and states the identifying assumptions. In Section 5, I present the main results and discuss the mechanisms. Section 6 outlines a cost-benefit analysis, and Section 7 concludes.

2 Background

2.1 Water Scarcity and Surface Irrigation System

South Africa is severely affected by water scarcity due to low average annual rainfall. The country receives only about 450 mm per year, much less than the world average of 860 mm (WRC, 2024). Furthermore, climate change is expected to exacerbate the situation by making droughts in southern Africa more frequent and severe (Chikoore and Jury, 2021).

Despite water scarcity, South Africa has developed a robust agricultural sector, largely thanks to irrigation that consumes 60% of available water resources. Irrigation systems cover 1.3 million hectares, accounting for 7.2% of arable land, and contribute to 30% of the country's crop production (Dennis and Nell, 2002). Irrigation systems can be classified

into three groups based on their components, cost, and performance: (1) flood irrigation, where water flows over the soil using gravity, (2) mobile systems, which irrigate as they move across fields (e.g., center-pivot systems), and (3) static systems, which remain stationary during irrigation (Reinders, 2011). Flood irrigation, though the least efficient and most prone to water loss, is the most affordable option. In contrast, the mobile and static systems, such as center pivot irrigation, are far more efficient but also the most capital-intensive method (Lichtenberg, 1989).

Irrigation in South Africa primarily relies on surface water.⁴ Surface water is released from large dams or weirs in rivers and is distributed via an extensive network of primary and secondary canals.⁵ There are four levels of water management infrastructure (Reinders et al., 2010). The first level is the “water source,” which stores water in a dam or reservoir. The second level is the “bulk conveyance system,” which transports water from the dam to several irrigation schemes via a canal. Alternatively, irrigation water can be extracted directly from a river. The third level is the “irrigation scheme” infrastructure, designed to deliver water from the main canal to the boundaries of participating farms. This includes on-scheme dams, canals, and pipes. The final level is the “irrigation farm” itself, where on-farm dams, canals, and irrigation systems (such as center pivots, sprinklers, and micro-sprays) are used to deliver water directly to the crops.

In this paper, when I refer to irrigation canals, I am specifically referring to the second level of water management infrastructure, which marks the beginning of the canal system (“canal heads”). At this point, water is released from a dam or weir into the main canal, which then carries it to irrigation schemes along its path. The distribution of water is controlled and monitored by water management authorities. Each user is entitled to the amount of water that they are registered for with the Department of Water and Sanitation. To understand water allocations, it is important to consider the history of water rights in South Africa.

2.2 History of Water Policy in South Africa

Water policy in South Africa has evolved over time, drawing its principles from Roman, Dutch, and later English legal traditions (Perret, 2002). The first notable legislation was the 1910 Irrigation Conservation of Water Act, which established the riparian principle as

⁴Groundwater irrigation accounts for only 1% of cultivated land (Altchenko and Villholth, 2015). Given that 7.2% of cultivated land is irrigated overall, this implies that about 14% of irrigated crops rely on groundwater, while the remaining 86% depend on surface water.

⁵Figure A–1 in the Appendix illustrates the number of large dams constructed per decade. The first large dam dedicated to irrigation was built in 1913, with the majority of dams completed by the early 1990s. Since then, only six large dams have been constructed, all completed in the 2000s.

the central feature. Under this principle, landowners whose properties bordered a water source had the right to extract water, with minimal state intervention.

The second major water legislation, the 1956 Water Act, reaffirmed the riparian principle. This system disproportionately benefited large-scale commercial farmers, primarily white landowners, who had unrestricted access to water. Under apartheid, non-white populations were excluded from land ownership and forcibly relocated to homelands, or Bantustans, where water access was limited.

After the end of apartheid, the 1998 National Water Act was introduced, discarding the riparian system and establishing the government as the custodian of water resources. This legislation formed new institutions for decentralized water management. The Department of Water and Sanitation was tasked with managing water resources, but most responsibilities were to be delegated to nineteen Catchment Management Agencies. Due to bureaucratic delays, only two agencies are fully operational today.⁶ Additionally, Water User Associations were created as cooperative groups of water users, allowing communities to pool financial resources to manage irrigation efficiently.

The primary goal of the 1998 National Water Act was to redress historical racial inequalities and ensure equitable water distribution to benefit all citizens. The Act introduced four types of water rights. First, water licenses represent a modern approach to water policy. They are granted by the Department of Water and Sanitation based on applications by individual users. Second, users who could demonstrate that their water usage was lawful during a two-year period before the Act's promulgation are allowed to retain their rights under the existing lawful water use provision. This category is commonly referred to as the "sunset clauses" and has faced criticism for effectively maintaining the status quo, making it difficult to redistribute water rights and move away from the inequalities entrenched during apartheid.⁷ Third, water use can be authorized under Schedule 1, which covers reasonable domestic water use that has minimal impact on water resources. Lastly, general authorizations permit water use without a formal license, within certain limits and conditions, often benefiting smallholder farmers with a relatively small impact on water resources. Water use is also subject to water tariffs that serve to recover operation and maintenance costs of water infrastructure and to manage water demand under scarcity. Commercial farmers were required to start paying these fees immediately after

⁶The only two operating agencies are the Breede-Gouritz and the Inkomati-Usutu Catchment Management Agencies. In the remaining parts of the country, the regional offices of the Department of Water and Sanitation take on water management responsibilities.

⁷For instance, see the meeting proceedings from the Water and Sanitation Committee of the National Assembly available at <https://pmg.org.za/committee-meeting/23583/>. [Accessed on 10 September 2024.] The perceived status quo was also the reason behind the proposal of racial quotas for water allocations by the Minister of Water and Sanitation on 19 May 2023. See, for instance, Bega (8 June 2023).

the law promulgation, while subsistence farmers were phased into the system over a period of five years (Speelman et al., 2009). Several policies have been implemented to help smallholder farmers with access to irrigation. More details on the history of smallholder irrigation schemes can be found in the Appendix.

In summary, there are three key takeaways from the background on irrigated agriculture in South Africa. First, irrigation plays a critical role for the country's agricultural productivity. Second, subsistence farmers were not entirely neglected in the development of irrigated agriculture. Smallholder irrigation schemes were implemented even during the apartheid era, although, the anecdotal evidence suggests that these schemes did not reach their potential. Moreover, subsistence farmers lacked secure water rights. Third, a major institutional shift occurred in the late 1990s following the end of apartheid. New water policy democratized access to water and there was a shift to a more decentralized management of water infrastructure. The central question that remains is how these institutional reforms have impacted both commercial and subsistence farmers in terms of their ability to benefit from irrigation resources.

3 Data and Summary Statistics

3.1 Data Sources

Agricultural outcomes. I estimate the effect of large-scale irrigation infrastructure on two agricultural outcomes: crop productivity and land use. Both outcomes are proxied by remote-sensing data. The main advantage of remote-sensing data is high granularity (Donaldson and Storeygard, 2016), which allows for an unambiguous definition of treated and untreated areas. An alternative source of data on agricultural outcomes are agricultural surveys. There are two main issues that make them unsuitable analysis. First, due to data privacy concerns the smallest geographical unit in publicly released datasets is often a province or a district, which would be too coarse for spatial RD.⁸ Second, agricultural surveys do not have universal coverage and even if I had access to GPS coordinates of each observation, it would be unlikely that there would be enough units above and below the running variable threshold. A disadvantage of using remotely sensed data, as opposed to an agricultural survey, is the inability to account for agricultural inputs, labor, or the size of the plot.

Agricultural yields are proxied by Enhanced Vegetation Index (EVI) derived from Land-

⁸For instance, the 2017 Census of Commercial Agriculture (Statistics SA, 2020) is available only at the province level. South Africa's Census of Agricultural Households is available at the level of district municipality.

sat 8 satellite imagery⁹ which is available at a 30-meter resolution. EVI is calculated from the Near-Infra-Red (NIR), Red and Blue image bands of each scene and ranges in value from -1 to 1, where negative values indicate non-vegetated areas such as water or barren land and positive values indicate dense vegetation. EVI improves on the Normalized Difference Vegetation Index (NDVI), which is another commonly used satellite-derived proxy for agricultural yields, by adjusting for canopy background and reducing atmosphere influences (Huete et al., 2002). It is therefore more suitable for areas with high amounts of biomass, such as irrigated land. Both EVI and NDVI have been shown to be reliable proxies for agricultural yields in various geographical contexts (Lobell et al., 2020; Burke and Lobell, 2017; Asher and Novosad, 2020). South Africa has two agricultural seasons and I define each season's measure of productivity as the maximum value of EVI over the growing and harvesting stages.¹⁰ The main crop of the summer season is maize which is grown and harvested from mid-November until mid-May. The main crop of the winter season is wheat which is grown and harvested from mid-June until the end of November.¹¹ South Africa also has two distinct climates in terms of when most of the rainfall occurs. West of the country is characterized by winter rainfall with smaller amounts of total precipitation, whereas the East is characterized by summer rainfall with larger amounts of precipitation.¹² This translates into distinct growing seasons for maize in the East and in the West. To ensure comparability of climate relevant to agricultural production, I focus my analysis on the Eastern part of the country.¹³

Data on land use comes from the South Africa National Land Cover 2018 (SANLC 2018) and the South Africa National Land Cover 1990/2020 Change (SANLC 1990/2020) released by the South African Department of Agriculture, Land Reform, and Rural Development. SANLC 2018 is a raster dataset available at the resolution of 20 meters, generated from automated mapping models using Sentinel 2 satellite imagery for the period of 1 January 2018 to 31 December 2018. SANLC 1990/2020 is based on a comparison of SANLC 1990 and SANLC 2020 datasets.¹⁴ I use the land cover data in two ways. First, I create a

⁹Landsat 8 Collection 1 Tier 1 8-Day EVI composite accessed through Google Earth Engine

¹⁰I am not concerned about contamination by non-crop vegetation because I restrict the analysis to agricultural land by applying a crop mask as discussed in the following paragraph.

¹¹See FAO Country Briefs for South Africa, <https://www.fao.org/giews/countrybrief/country.jsp?code=ZAF>. [Accessed on 15 September 2026.]

¹²Climate Change Knowledge Portal. South Africa. Current Climate. Climatology. <https://climateknowledgeportal.worldbank.org/country/south-africa/climate-data-historical>. [Accessed on 31 October 2023.]

¹³I focus on areas that lie east of the 25th meridian. This includes the provinces Limpopo, Mpumalanga, KwaZulu-Natal, Free State, Gauteng, Northwest, and parts of Northern Cape and Eastern Cape.

¹⁴The 72-class SANLC 1990 was generated from the Landsat 5 imagery for the period of 1989–1991 and is available at the resolution of 30 meters. The 73-class SANLC 2020 was generated from the Sentinel 2 imagery for the period of 1 January 2020 to 31 December 2020 and is available at the resolution of 20

crop mask and apply it to the data when analyzing agricultural productivity. This allows me to eliminate concerns about contamination of the productivity measure by non-crop vegetation. Second, I generate key outcomes to study the effect on agricultural production at the extensive margin and on land use change between 1990 and 2020 (pre- and post-apartheid).

Figure A-3 in the Appendix shows how remote-sensing data captures observable irrigation infrastructure and irrigated land. A raw satellite image is presented alongside the two principal datasets: SANLC 2018, with different colors representing different land cover classes, and the Landsat 8 EVI composite, with darker green denoting higher index values. The satellite image shows a dam with the head of the canal it feeds and the circular fields characteristic of center-pivot irrigation.¹⁵ SANLC 2018 classifies these circles as pivot-irrigated annual crops, and the EVI composite assigns them higher index values. The fact that two independent remote-sensing datasets identify the same land as irrigated supports the use of EVI as a proxy for crop yields.

Treatment status. To determine the treatment status I collect GPS coordinates of 144 canals located in the study region from the South Africa’s Department of Water and Sanitation¹⁶ and elevation data from ALOS Digital Surface Model (Tadono et al., 2014), available at a 30-meter resolution. First, I determine the elevation relative to the sea level for each of the 144 irrigation canals. Next, I determine the elevation of each grid cell in my dataset relative to the nearest canal by taking a difference between the elevation of the canal and the elevation of the given grid cell. Areas with positive relative elevation are defined as treated, whereas areas with negative relative elevation are considered as control.

Covariates. I gather several geophysical covariates to perform balance tests and verify the validity of RD estimates. These covariates are also included in the preferred specification in order to improve precision. Daily precipitation values are extracted at a 5,566-meter resolution from the Climate Hazards Center InfraRed Precipitation with Station (CHIRPS) dataset (Funk et al., 2015). I construct a precipitation measure as the total annual precipitation within each cell and then I average this value over the years 2014–2018. Next, I extract the daily land surface temperature at the resolution of 1,000 meters from the MODIS Terra Land Surface Temperature dataset (Wan, Hook and Hulley, 2021). I construct a tem-

peratures.

¹⁵Circular fields are characteristic of the pivot irrigation technology. A pivot is located in the center and rotates the irrigation equipment (an arm with sprinklers attached to it) around the field.

¹⁶The study region consists of areas near canals that lie east of the 25th meridian. The command area of some canals is located entirely within non-homelands, of others entirely within former homelands, while some canals’ command areas span both homelands and non-homelands. See Figure 1 to see the spatial distribution of canals and homelands.

perature measure as the monthly maximum temperature averaged over 2014–2018. The terrain ruggedness index (Nunn and Puga, 2012) measures topographic variability within a given area that quantifies the variation in elevation between neighboring cells. Finally, I calculate distance to the nearest river using WWF HydroSHEDS (Grill et al., 2019).

Homelands. Boundaries of former homelands were obtained from the Department of Agriculture, Land Reform, and Rural Development. Figure 1 shows how the spatial distribution of former homelands and the irrigation canals in the study region. Several irrigation canals are located inside former homelands or just outside, providing subsistence farmers with possible access to irrigation water.

South Africa 2011 Census. For the analysis of socio-economic outcomes, I use the 100% sample of the South Africa 2011 Census available at the small area level.¹⁷

3.2 Summary Statistics

The units of analysis are 30×30 m grid cells obtained by converting a raster dataset into a delimited text format. The sample is restricted to areas within ten kilometers of each canal head (56,465,817 grid cells). An additional bandwidth restriction on the running variable limits the sample to observations within 50 meters of relative elevation to the nearest canal head (29,049,656 grid cells). Among these observations, 6,795,394 are classified as agricultural land. Summary statistics for the full sample and the RD sample are presented in Table 1.

Restricting the sample to a narrow bandwidth of 50-meter relative elevation does not result in large differences compared to the full sample, although land in the RD sample is more likely to be agricultural. Columns (2) and (3) present differences between areas below and areas above canal heads, and columns (4) and (5) compare former homelands with non-homelands. Areas below canal heads have significantly higher values of EVI, they are more likely to be agricultural and to grow commercial irrigated annual crops. They are less likely to grow commercial rainfed annuals, be cultivated by subsistence farmers, or lie fallow. There are also significant differences in terms of geophysical covariates. Treated areas have slightly lower maximum monthly temperature (36.2 degrees versus 35.4 degrees) and slightly higher mean annual precipitation (523 mm versus 552 mm). They are also on average closer to the nearest canal (5,250 m versus 5,059 m), and to the nearest river (1,110 m versus 836 m). Although these differences are significant, they do not pose a concern for the RD identification strategy since the key RD assumption demands absence of discontinuity at the threshold, and not the same average values of covariates at

¹⁷A small area is an aggregation of enumeration areas (EAs) and it is the most granular unit made publicly available.

either side of the threshold. The assumption on the absence of discontinuity in covariates is tested in Table 2.

Former homelands cover 9% of the RD sample. They tend to be slightly more agricultural than non-homelands (24.1% versus 23.3%). Sixty-eight percent of agricultural land in former homelands is cultivated by subsistence farmers and 30% lies fallow. Only 1% of homelands agricultural land is exploited by commercial farmers. In terms of geophysical characteristics, homelands tend to have higher maximum monthly temperature (36.5 degrees versus 35.9 degrees), higher mean annual rainfall (579 mm versus 528 mm), they are on average further away from the canals (6,154 m versus 5,129 m) and slightly closer to rivers (1,016 m versus 1,029 m). They also lie at a lower elevation above the sea level (846 m versus 1,109 m).

4 Empirical Strategy

4.1 Previous Literature

The main challenge in estimating the impact of large-scale irrigation infrastructure is their endogenous placement. For instance, in order to maximize the return on investment, irrigation dams (and the corresponding network of irrigation canals) might be placed in areas with better agricultural potential. Alternatively, irrigation dams might be constructed in politically favored places, which could be correlated with the provision of other goods and services correlated with agricultural yields. A related issue is the lack of historical data for the pre-construction period, which prevents the use of panel data methods. Most canals were built in the 1960s and 1970s and the Landsat 8 EVI measures are available only from 1984 onward.

The previous literature employed two main strategies to deal with this endogeneity issue. First, early studies relied on the instrumental variable approach. [Duflo and Pande \(2007\)](#) use river gradient which reflects geographic suitability to predict the placement of irrigation dams across India. A similar approach was applied also in the context of sub-Saharan Africa. In addition to the river gradient, [Strobl and Strobl \(2011\)](#) use ephemeral and perennial river type as an instrument to predict the distribution of dams across the continent.

Second, the RD approach recently gained popularity as a result of the increasing availability of high-resolution data on crop yields. The running variable is either the distance to the command area boundary ([Blakeslee et al., 2023](#); [Jones et al., 2022](#); [Hagerty, 2022](#)) or the relative elevation to the nearest canal ([Asher et al., 2024](#)). A key advantage of RD design is that it provides causal estimates under weaker assumptions compared to the

IV approach. The main assumption for the internal validity of RD estimates is the continuity at the threshold of all the observable and unobservable characteristics that could be correlated with the outcome. In contrast, the IV method requires the assumptions of relevance, exogeneity, and the exclusion restriction, which are more likely to be violated. For example, the river gradient used as an instrument in [Duflo and Pande \(2007\)](#) might influence crop yields directly and not only by increasing the probability of irrigation dam presence. While lower river gradient makes the area more suitable for dam construction, it is also mechanically correlated with the land gradient, which makes the area more suitable for agriculture and generates higher yields. Such correlation would violate the exclusion restriction. Additionally, the IV approach estimates only the local average treatment effect (LATE) for compliers, that is, the impact of only those dams that were built due to favorable geophysical conditions. It does not provide insights into the effects of dams built for other reasons, making the RD approach more appealing in this context. However, the caveat that RD estimates capture only local treatment effects remains, as the estimates apply only to observations within the selected bandwidth around the cutoff of the running variable.

In this paper I use an RD design with relative elevation to the nearest canal head as the running variable. Figure [A-2](#) in the Appendix provides a visual representation of the elevation-based RD design. My preferred specification regresses each outcome on a treatment indicator and a linear function of the running variable while allowing the slope to differ between the treatment and control groups.

4.2 Regression Discontinuity Design

The empirical strategy relies on the fact that surface irrigation is mostly gravity-based. Water from either large irrigation dams or from weirs in rivers is distributed through a canal that carries water toward irrigation schemes located in downhill areas.¹⁸ Because it is technically possible to pump water uphill from the main canals, the running variable does not perfectly predict the actual treatment status. However, it is cheaper to deliver water downhill, and therefore, land below canals has a higher probability of being irrigated.

The RD specification estimates the local average treatment effect (LATE) of large-scale irrigation infrastructure on agricultural productivity and land use outcomes for areas just below canal heads. Following [Imbens and Lemieux \(2008\)](#) and [Gelman and Imbens \(2019\)](#), I regress each outcome on the treatment indicator (equal to one if an observation lies below the canal head), a linear function of the running variable, and an interaction between the treatment indicator and the running variable:

¹⁸See Section 2 for detailed information on surface irrigation infrastructure.

$$Y_{ic} = \beta_0 + \beta_1 Treat_{ic} + \beta_2 Rel_Elev_{ic} + \beta_3 Rel_Elev_{ic} \times Treat_{ic} + \beta_4 X_{ic} + \mu_c + \epsilon_{ic} \quad (1)$$

Y_{ic} is the outcome of interest in grid cell i near canal c . $Treat_{ic}$ is the treatment indicator that is equal to one when a grid cell lies below a canal head. The running variable Rel_Elev_{ic} is calculated as the difference between the elevation of grid cell i and the elevation of canal c .¹⁹ Observations with negative values of Rel_Elev_{ic} lie below a canal head (treatment group) and observations with positive values lie above a canal head (control group). The running variable Rel_Elev_{ic} is interacted with the treatment indicator to allow for different slopes at each side of the threshold.

X_{ic} refers to five geophysical covariates that are controlled for to improve precision: average maximum monthly temperature, average total annual precipitation, terrain ruggedness index, distance to the nearest canal head, and distance to the nearest river. Finally, canal fixed effects μ_c are included to account for unobserved, time-invariant characteristics that may vary across canals, such as differences in agricultural policy, institutions or other infrastructure, that may affect the outcome variable.

Standard errors are clustered at the canal level since it is the unit at which the treatment is assigned (Abadie et al., 2023). This also allows for arbitrary spatial correlation in places around each canal. To ensure comparability of treatment and control units, I restrict the sample to grid cells that lie within ten kilometers of distance and fifty meters of relative elevation to the nearest canal.

4.3 Identifying Assumptions

The treatment effect is identified as the magnitude of a discontinuity in the outcome variable estimated at the zero relative elevation threshold: $\lim_{x \rightarrow 0^-} \mathbb{E}[Y_{ic} | Rel_Elev_{ic} = x] - \lim_{x \rightarrow 0^+} \mathbb{E}[Y_{ic} | Rel_Elev_{ic} = x]$. Two key assumptions are needed to ensure causal interpretation of the RD estimates.

First, the treatment must be at least partially determined by the running variable, implying that land below a canal should have a higher likelihood of being irrigated than land above. Mathematically, $\lim_{x \rightarrow 0^-} \mathbb{E}[Irrig_{ic} | Rel_Elev_{ic} = x] \neq \lim_{x \rightarrow 0^+} \mathbb{E}[Irrig_{ic} | Rel_Elev_{ic} = x]$ where $Irrig_{ic}$ denotes irrigation status of grid cell i . Figure 2 and Table 3 (columns (1) and (2)) shows the probability of land growing irrigated annual crops as a function of the running variable in non-homelands. In former homelands the first stage cannot be

¹⁹Elevation of each grid cell comes from the ALOS Digital Surface Model, available at 30-meter resolution. This matches the resolution of the EVI outcome variable and is close to that of the land use outcomes (20 meters), so treatment status is assigned at the grid-cell rather than a more aggregate level.

estimated, because SANLC 2018 does not distinguish irrigated from non-irrigated subsistence land. This does not threaten identification but it does change the interpretation: the homeland estimates are reduced-form effects of being located below a canal head rather than effects of having access to irrigation.

Panel (a) of Figure 2 shows the unconditional probability of land being classified in SANLC 2018 as growing irrigated annual crops (maize or wheat). The estimated discontinuity at the zero relative elevation threshold is 0.0227 (standard error = 0.0071), which is statistically significant at the 1% level. In areas above canals, 4.2% of all grid cells are classified as growing irrigated annuals. Land below canals is 2.27 percentage points more likely to grow irrigated annuals, which represents a 54% increase relative to the above-canal control group mean. Because treatment assignment does not change sharply at the cutoff, the RD specification therefore follows a fuzzy design. Furthermore, the slope of the first stage variable changes at the cutoff value. This pattern reflects the fact that commercial farmers in non-homelands pump irrigation water uphill from the canals. For areas above a canal, the cost of pumping increases with vertical distance, as indicated by the rising probability of irrigation as relative elevation approaches the zero relative elevation threshold. In contrast, for areas below a canal, the absence of pumping costs results in a relatively flat relationship between vertical distance and the probability of irrigation.

Panel (b) of Figure 2 shows the probability of land growing irrigated annuals in non-homelands, conditional on the land being agricultural. The estimated discontinuity at the zero relative elevation threshold is 0.0182 (standard error = 0.0138). This effect is no longer statistically significant. In areas above canals, 19.9% of agricultural grid cells are classified as growing irrigated annuals, while the remaining 80.1% grow rainfed annuals or are left fallow. Agricultural land below canals is 1.82 percentage points more likely to grow irrigated annuals, which represents a 9.15% increase relative to the mean of the above-canal control group. The conditional first-stage effect is smaller in magnitude than the unconditional one, which suggests that the unconditional effect largely operates through the extensive margin. Canals induce some marginal land that would have remained non-agricultural to be brought into farming as irrigated cropland. Again, this does not threaten identification but it may affect the interpretation of the main estimates, which would capture the reduced-form effects of being located below a canal head rather than causal effects of irrigation.

Second, all the observable and unobservable variables that also determine the outcome must be continuous at the threshold. In other words, there should be no discontinuity in any characteristics that could also affect yields between places that are just above and just below a canal head. Mathematically, $\lim_{x \rightarrow 0^-} \mathbb{E}[X_{ic} | Rel_Elev_{ic} = x] =$

$\lim_{x \rightarrow 0^+} \mathbb{E}[X_{ic} | Rel_Elev_{ic} = x]$ for all observable and unobservable characteristics X_{ic} . The only source of discontinuity must be access to irrigation due to the presence of the canals. Note that the purpose is to estimate the *long-run* effect of irrigation canals. In the long-run, farmers might adjust their input use to take into account the increase in water supply due to irrigation (Hagerty, 2022). Therefore, a discontinuity at the threshold in, for instance, the use of fertilizer would not be a threat to identification, but rather one of the components of the long-run effect. In order to better articulate this concept, suppose that crop yields $Y(W, I)$ are a function of water supply W and other inputs I (fertilizer, pesticides, labor, etc.). In the long-run, inputs are assumed to be adjusted to the level of available water supply. The effect of water supply on crop yields is then given by the total derivative of $Y(W, I)$ with respect to W which consists of a direct effect and an indirect effect:

$$\frac{dY(W, I)}{dW} = \underbrace{\frac{\partial Y(W, I)}{\partial W}}_{\text{direct effect}} + \underbrace{\frac{\partial Y}{\partial I} \frac{dI(W)}{dW}}_{\text{indirect effect}} \quad (2)$$

In this paper, I estimate the total effect which includes both the direct and indirect effects.

To verify the smoothness assumption, I examine five geophysical characteristics (average maximum monthly temperature, average total annual precipitation, terrain ruggedness index, distance to the nearest canal, and distance to the nearest river) and three variables related to soil quality (proportions of sand, silt, and clay particles). Figures 3 and 4 present binned scatter plots for these covariates, residualized on canal fixed effects. Table 2 reports statistical tests for discontinuities at the cutoff, estimated using Equation 1 on grid cells within 10 kilometers of a canal and within 50 meters of relative elevation. In both the agricultural and full samples, and for both homelands and non-homelands, I find no large or statistically significant discontinuities in terrain ruggedness, temperature, precipitation, and soil quality. However, Table 2 reports statistically significant discontinuities in distance to the nearest canal and distance to the nearest river. Treated areas immediately below the cutoff are located farther away from the nearest canal and the nearest river. Both could plausibly arise from the logic of canal construction combined with the bandwidth restriction on relative elevation. If canals are built to serve the largest area possible, then land below may extend farther away while remaining within the 50-meter vertical distance bandwidth. Above the canal, cells at comparable horizontal distances are more likely to exceed the 50-meter bandwidth, and therefore are excluded from the RD sample, mechanically generating larger distance to the nearest canal for areas below. Similarly, canals may be constructed so as to serve areas that are farther from rivers and therefore less likely to have access to river-based irrigation. Additionally, the discontinuities in both distance variables have the same sign across homelands and non-homelands,

making it less likely that they could account for the heterogeneous treatment effects. I control for distance to the nearest canal, distance to the nearest river, and other weather and terrain covariates in the main specification. Soil quality variables are controlled for only as a robustness check because of a high number of missing values.

5 Results

5.1 Agricultural Seasons in South Africa

South Africa has two distinct agricultural seasons during which winter and summer crops are grown. Winter crops are typically planted during the autumn months of April and May, and then harvested during the winter and early spring months of September to November. Wheat is the most important winter crop with a production of 2,263,000 tonnes during the 2021/2022 season, although most production is concentrated in Western Cape²⁰

Summer crops are typically planted in the early summer months of October to December, and are harvested during the late summer and early autumn months of February to April. The most important summer crop is maize with a production of 15,810,000 tonnes during the 2021/2022 season (SAGIS, Monthly Producer Deliveries). Maize is produced in almost all parts of South Africa, but the majority of production is concentrated in the Free State, Mpumalanga, North West, and KwaZulu-Natal provinces. I use the temporal ranges of growing and harvesting seasons of wheat and maize (east) to construct the EVI measures of agricultural productivity for the wheat and maize seasons.

5.2 The Intensive Margin Effects: Canals Increase Crop Yields in Non-Homelands

First, I examine the effects of irrigation canals on the intensive margin of agricultural production. Table 3 reports the results of estimating Equation 1, separately for non-homelands (Panel A) and former homelands (Panel B). Binned scatter plots are presented in Figure 5. I restrict the analysis to grid cells identified as agricultural to ensure that results are not contaminated by non-crop vegetation.

In non-homelands, I find that irrigation canals increase agricultural productivity by 3.9% during the wheat season and by 2.3% during the maize season (columns (3) and (4)). Note that these are reduced-form estimates and are not scaled by the discontinuous change in probability of land being irrigated at the cutoff.

²⁰South African Grain Information Service (SAGIS), *Monthly Producer Deliveries per Province*, released 27 March 2023: <https://www.sagis.org.za/monthly-prod-deliveries.html>. [Accessed on 8 January 2024.]

In former homelands, where mostly subsistence farmers operate (as established in Table 1), irrigation canals do not have the same effects. For both the maize and wheat seasons, the estimates are negative, though the wheat season results are not statistically significant. There are two possible explanations for this outcome. Either subsistence farmers are entirely excluded from water institutions and therefore lack access to water from large-scale irrigation schemes, or they are granted water use licenses but are unable to extract and productively use sufficient quantities due to market failures preventing them from building and maintaining last-mile infrastructure.

5.3 The Extensive Margin Effects: Canals Increase the Area Under Production in Non-Homelands

Next, I consider how irrigation canals affect the extensive margin of agricultural production. In other words, do irrigation canals increase the area under cultivation? To answer this question, I estimate Equation 1 on the full RD sample, *without* restricting the analysis to agricultural land. As before, the specification is estimated separately for non-homelands (Panel A) and former homelands (Panel B).

The results are reported in Table 3 and represented visually in Figure 6. In non-homelands, areas below canals have a 20% higher probability of being cultivated with annual crops (an increase of 4.3 pp relative to the mean in the control group of 21.2%). In contrast, land below canals in former homelands is not more likely to be cultivated, although it is 37% less likely to lie fallow (a decrease of 3.3 pp relative to the mean in the control group of 9%). There is no statistically significant effect on the likelihood of fallow land in non-homelands. These findings are consistent with the previously discussed crop yield effects. If subsistence farmers located downstream from irrigation canals do not benefit from higher yields, then they may be less motivated to expand area under production. The presence of canals does not increase the probability of cultivation of perennial crops.

5.4 The Extensive Margin Effect Can Be Explained by the Post-Apartheid Land Use Changes

A potential concern with the results in the previous section is that the discontinuity in cultivated area may simply reflect pre-existing differences in agricultural land use rather than the causal effect of access to irrigation. Large dams and canals could have been built to serve existing fields and maximize the amount of area cultivated downstream. Ideally, I should examine the RD effects before the construction of the irrigation infrastructure to test whether the probability of annual crops being grown was the same above and below

the future canal head. However, historical land-use data are unavailable for the period preceding the completion of most large-scale irrigation projects.²¹

An alternative strategy is to consider the change in land use outcomes between 1990 and 2020. Before the 1994 end of apartheid water access was based on riparian rights, which mean that landowners had the right to extract as much water as they wanted from water resources adjacent to their property. This would often leave downstream farmers with not enough water. Additionally, non-white population was excluded from land ownership and lacked secure water rights. The post-apartheid water legislation aimed to democratize access to water with a particular goal of redressing past racial inequalities and introduced a system of water licensing. Therefore, we should expect to see an expansion of cultivated land below canals as more farmers are granted water rights.

Table 4 and binned scatter plots in Figure 7 show the estimated RD effects on the expansion of agricultural land between 1990 and 2020, separately for homelands and non-homelands. I focus on four types of land expansion: any agricultural land growing annual crops, commercial pivot irrigated land, commercial non-pivot irrigated or non-irrigated land,²² and subsistence land. In non-homelands, treated areas below canals experienced a 27.8% higher expansion of agricultural land with annual crops than comparable areas just above canals (an increase of 1 pp relative to the control group mean of 3.5%). There was no such expansion in former homelands. Irrigation canals, therefore, contributed to the expansion of the commercial agriculture in South Africa but did not affect subsistence farmers.

5.5 Robustness Checks

I perform several robustness checks to test the sensitivity of my main results to different empirical choices.

Extending the analysis to multiple years. A potential concern is the temporal external validity of the effects on crop yields, as I only use data for one year. It is possible that the negative effects of irrigation canals on subsistence farmers would not be observed in other years. In particular, 2018 was unusual due to a country-wide drought, and it could be that subsistence farmers situated below canals are negatively impacted only during droughts.²³

²¹Most large irrigation dams were completed by the 1960s and the 1970s. See Figure A-1 in the Appendix.

²²These two categories of commercial land are combined because SANLC 1990 cannot distinguish between irrigated and non-irrigated land without the visual cue of the circular fields. As explained earlier, center pivot irrigation results in easily distinguishable circular patterns and that's why SANLC 1990 can still distinguish pivot irrigated land.

²³During droughts, there is typically less water available in the irrigation system. Commercial farmers might be prioritized over subsistence farmers for access to the limited water supplies, which could lead to lower yields for subsistence farmers below canals compared to those above canals as they might not be able

To address this issue, I extend the analysis to multiple years.

To identify agricultural land, the main results rely on SANLC 2018, which covers only one year. To extend the analysis to multiple years, I generate a new crop mask based on the Global Cropland Maps produced by the University of Maryland Global Land Analysis & Discovery (UMD GLAD) for the period 2000–2019 (Potapov et al., 2022). I use all five UMD GLAD datasets, each covering a four-year period.²⁴ When I compare the crop masks generated by UMD GLAD 2019 and SANLC 2018, I find them to be largely consistent.²⁵ Applying the extended crop mask, I estimate an equation similar to 1, with the main difference that observations are indexed by year and the regressions include year fixed effects.

The results reported in Table A–1 confirm that the negative effects of irrigation canals in former homelands remain in other years as well. In non-homelands, irrigation canals continue to have a positive effect on yields, although the magnitude is smaller and the coefficient is not statistically significant for maize. This could be the case because the UMD GLAD crop mask includes perennial crops, which attenuates the estimated effect since perennial crops exhibit little to no seasonal variation in biomass.

Different vertical bandwidths. The results in Tables A–3 and A–4 show robustness to different vertical bandwidths of four main outcomes (log wheat EVI, log maize EVI, the probability of land being agricultural, and the probability of annual crops being grown). To ensure comparability of treated and untreated land, my preferred specification restricts the sample to grid cells that are within 50 m of relative elevation to the nearest canal.

In Tables A–3 and A–4 I change the vertical bandwidth to 25 m, 75 m, and 100 m. The wider bandwidths lead to RD estimates that are larger in magnitude and highly statistically significant. In the former homelands, the 25 m bandwidth also generates estimates consistent with the previous results. On the other hand, in non-homelands, the 25-meter bandwidth generates smaller and statistically insignificant estimates. This could be explained by high access to irrigation on both sides of the RD cutoff. It is possible that most

to easily adjust to the restricted water supply.

²⁴I use UMD GLAD 2000, 2004, 2007, 2011, 2015, and 2019. For example, UMD GLAD 2015 classifies a grid cell as cropland if active crops were detected during the years 2012–2015. Potapov et al. (2022) define cropland as “land used for annual and perennial herbaceous crops for human consumption, forage (including hay), and biofuel.” (p.19)

²⁵In non-homelands, 87% of grid cells classified as cropland by UMD GLAD 2019 are also classified as agricultural in my previous analysis using SANLC 2018. In homelands, only 37% of grid cells classified as cropland by UMD GLAD 2019 are also classified as agricultural in my previous analysis. This discrepancy is mainly due to the presence of sugarcane, which is correctly detected as cropland by UMD GLAD but is not included in my previous analysis, where I focus only on annual crops. For a more detailed comparison of the SANLC and UMD GLAD datasets, Table A–2 in the Appendix provides a breakdown of NLC categories for both homeland and non-homeland detected cropland.

commercial farmers up to 25 meters above canals are able to pump water uphill from the canal distribution point to their fields. This would result in no visible differences in yields.

Different horizontal bandwidths. Tables A-5 and A-6 display robustness to decreasing the horizontal distance to the nearest canal. After restricting the distance to 2,500 m, 5,000 m, and 7,500 m, the results remain similar to those of the preferred specification, although the crop yield effects lose statistical significance in the former homelands, which is most likely due to the lower sample sizes.

Excluding a donut hole. As previously mentioned, commercial farmers located just above irrigation canals can pump water uphill from the distribution point to their fields. A more appropriate empirical approach might be a donut RD, where observations right around the cutoff are excluded (Barreca et al., 2011). Additionally, the donut RD approach could help mitigate measurement error bias that might otherwise attenuate the main results.²⁶ The donut RD results reported in Tables A-7 and A-8. As the size of the donut hole increases, the RD estimates also increase in magnitude, but the main finding remains consistent: commercial farmers in non-homelands benefit from higher yields and land expansion, whereas subsistence farmers do not. While the larger estimates may be due to the reduction of measurement error and the consequent attenuation bias, recent literature suggests that donut RD estimates could be biased under the standard RD assumptions (Noack and Rothe, 2023). Therefore, I present these results only as a robustness check.

Controlling for soil quality and different level of fixed effects. Finally, I check that the main results are not biased by the endogenous placement of canals so as to benefit land with better (or worse) soil quality. Even though I found no discontinuity in the soil quality variables around the cutoff (Table 2), I include them as covariates in the RD regressions to verify that this would not affect the main estimates. In Tables A-9 and A-10, I control for additional soil quality indicators, including the proportions of sand, silt, and clay particles in the fine earth fraction, which determine the type of soil, as well as soil carbon and nitrogen content.²⁷ The results remain qualitatively unchanged. The reason for not including soil quality variables in the main specification is the high number of missing values. I also present findings from models that exclude any covariates and from those that adjust the level of fixed effects from canals to local municipalities.

²⁶The measurement error could come from an imprecise determination of elevation of grid cells. Grid cells that have approximately the same elevation as the nearest canal might be assigned to the wrong treatment status. This type of measurement error is particularly likely because the reported vertical accuracy (the possible height difference between the modeled height and the actual height) of the ALOS DEM is 5 meters.

²⁷I include the sand, silt, and clay particles and the carbon and nitrogen content as two separate sets of soil control variables since the carbon and nitrogen are potentially endogenous to the presence of irrigation and might bias the results.

5.6 Local Labor Markets: Canals Decrease Unemployment in Non-Homelands

So far, the results show that large-scale irrigation infrastructure increases crop yields and expands area under cultivation for commercial farmers located in non-homelands. But do the benefits from irrigation extend beyond these commercial landowners? It is possible that the expansion of commercial farms benefits the rural poor through job creation and thus contributes to a decrease in rural poverty.

To examine the effects of irrigation infrastructure on local labor markets, I use a similar RD strategy. The main difference is the unit of analysis since the labor-market outcomes constructed from census data are available at a coarser spatial level. This creates a challenge of assigning treatment status to each census enumeration unit (small area), which is substantially larger than the 30×30 m grid cells used in the analysis above. To address this, I follow the approach of [Asher et al. \(2024\)](#) and define treatment status based on the fifth percentile of the elevation distribution within each small area. Specifically, I compare this percentile to the elevation of the nearest canal, assigning a small area to the treatment group if its fifth-percentile elevation lies below the elevation of the canal. I then estimate the following equation:

$$Y_{ac} = \alpha_0 + \alpha_1 Treat_{ac} + \alpha_2 Rel_Elev_{ac} + \alpha_3 Rel_Elev_{ac} \times Treat_{ac} + \mu_c + \epsilon_{ac} \quad (3)$$

Y_{ac} is the outcome variable averaged over households living in small area a near canal c , $Treat_{ac}$ is the treatment indicator equal to one if the fifth percentile of the distribution of elevation grid cells constituting a small area is less than the elevation of the nearest canal, and Rel_Elev_{ac} is the relative elevation to the nearest canal. The regression includes canal fixed effects and standard errors are clustered at the canal level. The sample is restricted to small areas that lie within 10 km of horizontal distance and 50 m of vertical distance to the nearest canal. α_3 is the coefficient of interest.

Table 5 shows the RD estimates of the effect of irrigation canals on employment and household income. Binned scatter plots are presented in Figure 8. In non-homelands, the unemployment rates decrease by 4.9 pp (20% relative to the mean of the control group) and the fraction of households with zero income decreases by 2.7 percentage points (19% relative to the mean of the control group). However, the results are significant only in non-homelands. In former homelands, unemployment and the proportion of households with zero income is also lower (by 5% and 6% respectively), but the effects are much smaller in magnitude and not statistically significant. Furthermore, irrigation canals have an imprecisely estimated positive effect on the average annual household income per capita and

the average annual household income per capita among the bottom 10% of the income distribution.

These results suggest that commercial farms in non-homelands generate new employment opportunities for the rural poor. Therefore, large-scale irrigation investments appear to contribute to poverty reduction, which is in line with previous findings on irrigation access in general (Duflo and Pande, 2007; Dillon, 2011; Sellamuttu et al., 2014; Sekhri, 2014; Boudot-Reddy and Butler, 2025; Gebregziabher, Namara and Holden, 2009; Dyer and Shapiro, 2023). The link between employment and poverty is not necessarily straightforward since the quality of jobs matters. If the jobs generated by the irrigation infrastructure are stable and well-paying, this poverty reduction is likely to follow. However, if the jobs are seasonal and low-paying, the resulting effects on poverty might be more limited. Nonetheless, both the reduction in unemployment and the decrease in the proportion of zero-income households are nontrivial indicators of the positive welfare effects of large-scale irrigation infrastructure.

To further understand the overall welfare impacts of large-scale irrigation infrastructure, I also conduct a cost-benefit analysis (Section 6). While the reduction in unemployment and its potential link to poverty alleviation is an important indicator of the benefits of this type of infrastructure, a cost-benefit analysis provides a more comprehensive evaluation by comparing direct economic benefits (higher agricultural productivity and increased area under production) to direct costs of construction. This method can help provide a clearer picture of whether the benefits of these irrigation projects outweigh their costs.

5.7 Mechanisms: Why Irrigation Infrastructure Fails to Benefit Subsistence Farmers in Former Homelands

Farmers in former homelands do not benefit from irrigation canals in terms of crop yields and agricultural land expansion. Moreover, for maize, canal exposure is associated with a statistically significant reduction in yields. I now provide suggestive evidence on the mechanisms behind the lack of agricultural productivity improvements in former homelands. First, as is well-documented in the literature on agricultural technology adoption, multiple constraints prevent small-scale farmers in sub-Saharan Africa from adopting profitable technologies (Suri and Udry, 2022; Suri et al., 2024), including irrigation (Jones et al., 2022). I highlight two potential mechanisms behind these null (or negative) effects for subsistence farmers: (1) capital constraints leading to the absence of last-mile infrastructure connecting farmers to the main irrigation canals and (2) water diversion from rivers downstream from the canals.

5.7.1 Capital Constraints and Absence of Last-Mile Infrastructure

Evidence from heterogeneous treatment effects by distance to canal head. Having agricultural land located topographically below an irrigation canal does not automatically result in access to irrigation. Effective water delivery requires a network of secondary canals that distribute water from the main canal to individual plots. In South Africa, the operation and maintenance of this secondary irrigation infrastructure is the responsibility of Water User Associations (WUAs), established under the 1998 National Water Act, and replacing the apartheid-era irrigation boards.²⁸ WUAs operate at a local level and are composed of individual water users. It is possible that WUAs in former homelands face larger capital constraints than WUAs in non-homelands, restricting their ability to construct and effectively maintain last-mile infrastructure connecting main canals to members' fields. If this mechanism is relevant, treatment effects should vary with distance to canal heads in former homelands, but not in non-homelands. Subsistence farmers farther away from the starting point of a canal would receive fewer benefits than farmers located closer to it. In contrast, if WUAs in non-homeland areas face fewer capital constraints and maintain more effective distribution systems, treatment effects should not systematically decline with distance. To test this hypothesis, I interact the treatment indicator with distance to the canal head ($distance_{ic}$) and estimate the following specification separately for former homelands and non-homelands:²⁹

$$Y_{ic} = \gamma_0 + \gamma_1 Treat_{ic} + \gamma_2 Treat_{ic} \times distance_{ic} + \gamma_3 Rel_Elev_{ic} + \gamma_4 Rel_Elev_{ic} \times Treat_{ic} + \gamma_5 X_{ic} + \mu_c + \epsilon_{ic} \quad (4)$$

The results are reported in Table 6, columns (1) and (3). In non-homelands (Panel A), the positive treatment effect of irrigation canals does not vary with distance to the canal head. The coefficient on the interaction term between treatment and distance is small and statistically insignificant, implying that farmers below canals who are located farther away benefit to a similar extent as farmers who are located nearby. Distance to the canal starting point therefore does not appear to constrain access to irrigation water. On the other hand, distance to the canal head matters for the size and direction of the treatment effect in former homelands (Panel B). The coefficient on the treatment indicator becomes positive, while the coefficient on the interaction term between treatment and distance to the canal head is negative and statistically significant. This pattern implies that the benefits

²⁸Department of Water and Sanitation (1998) National Water Act: Chapter 8: Water user associations. Available at: https://www.dws.gov.za/iwqs/nwa/tmp_Chapter_8.html [Accessed: 11 February 2026.]

²⁹Distance to the canal head is included in the vector of covariates X_{ic} .

of canal exposure decline with distance and eventually turn into losses. At the canal head (distance equal to zero), being located below canals is associated with a 2.7% increase in wheat EVI (not statistically significant) and a 6% increase in maize EVI (significant at the 10% level). At one kilometer from the canal head, the implied treatment effects decrease to 1.8% and 4.5% for wheat and maize respectively.³⁰ At ten kilometers, treatment effects turn negative: -6.1% for wheat and -7.9% for maize. If the costs of constructing and maintaining secondary canal networks increase with distance from the canal head, these findings are consistent with capital constraints limiting irrigation water delivery in former homelands.

5.7.2 Water Diversion from Rivers

Evidence from heterogeneous treatment effects by distance to nearest river. While capital constraints can explain null effects of canal exposure in former homelands, they cannot explain negative effects. When irrigation canals are built, they usually result in reduced flow in rivers located downstream from the canal head. If subsistence farmers rely primarily on rivers as a source of irrigation water, the canal-induced diversion of river water could be a mechanism behind the negative effects that these farmers experience. If this mechanism is relevant, treatment effects should vary with distance to nearest river in former homelands, but not in non-homelands. The test of this hypothesis is analogous to the previous test of the capital constraints mechanism. I interact the treatment indicator with distance to nearest river ($dist_river_{ic}$) and estimate the following specification separately for homelands and non-homelands:³¹

$$Y_{ic} = \delta_0 + \delta_1 Treat_{ic} + \delta_2 Treat_{ic} \times dist_river_{ic} + \delta_3 Rel_Elev_{ic} + \delta_4 Rel_Elev_{ic} \times Treat_{ic} + \delta_5 X_{ic} + \mu_c + \epsilon_{ic} \quad (5)$$

Columns (2) and (4) in Table 6 show the results. As in the case of capital constraints, the pattern differs sharply between non-homelands and former homelands. In non-homelands, the treatment effect does not vary with distance to the nearest river. The coefficient on the interaction term is small and statistically insignificant for both wheat and maize (columns (2) and (4) in Panel A), implying that commercial farmers' access to irrigation is unaffected by their proximity to rivers. This is consistent with commercial farmers obtaining water primarily through the canal network rather than from rivers.

³⁰The effects are calculated as $(\exp(\gamma_1 + \gamma_2 \times distance) - 1) \times 100$.

³¹Distance to nearest river is included in the vector of covariates X_{ic} .

In former homelands, the interaction term is positive and statistically significant for both wheat and maize (columns (2) and (4) in Panel B). This implies that the negative effect of canal exposure is concentrated among subsistence farmers who live in close proximity to rivers, while farmers located farther from rivers are relatively unaffected because they are less likely to rely on rivers for irrigation. The implied treatment effect for a farmer located directly on a riverbank (distance to nearest river equal to zero) is -4.6% for wheat (not statistically significant) and -4.3% for maize (statistically significant). As distance to the nearest river increases, the negative effect diminishes, consistent with water diversion being less binding for farmers who were not relying on river water for irrigation in the first place. At one kilometer away, the magnitude of the negative treatment effect declines to -2.3% for wheat and -1.9% for maize.³² At four kilometers away, the implied treatment effects turn positive: +5% for wheat and +5.8% for maize.

Evidence from a water user database. The positive coefficient on the interaction term between the treatment indicator and distance to nearest river is suggestive evidence that subsistence farmers in former homelands are negatively affected by the presence of irrigation canals due to the water diversion mechanism. To further support this mechanism, I leverage data from the South African Water Use Authorization & Registration Management System (WARMS) to show that subsistence farmers in former homelands indeed rely on rivers as their primary irrigation source.³³

Table 7 reports the breakdown of sources of water for farmers who registered for water use for the purpose of irrigated agriculture, separately for homelands and non-homelands. In both homelands and non-homelands, the dominant source of irrigation water are rivers. However, the share of farmers who use river water for irrigation is much higher in former homelands: 60.8% vs. 38.6%. In former homelands, rivers crowd out schemes as potential irrigation water source. While in non-homelands, 25.7% of irrigators use water from surface irrigation schemes, only 8.6% of farmers in former homelands do so. Farmers in former homelands relying on rivers for irrigation constitutes further evidence in support of the water diversion mechanism. Lastly, using river runoff data, I demonstrate that canals do divert water from rivers during peak irrigation season.

Evidence from river runoff data. Subsistence farmers in former homelands rely on rivers as a source of irrigation water. Farmers located below canals and near rivers can experience lower agricultural productivity, if canals divert water away from downstream

³²The effects are calculated as $(\exp(\delta_1 + \delta_2 \times \text{dist_river}) - 1) \times 100$.

³³The WARMS data is a register of all water users in South Africa as of 2024. I thank the Department of Water and Sanitation for kindly sharing the WARMS data with me.

river segments, particularly during the peak irrigation season. To examine this mechanism, I compile data on monthly river flow from the Global Runoff Data Center (GRDC), measured at 451 river monitoring stations across South Africa.³⁴ I use this information to show that canals divert water from upstream segments, affecting downstream river flow.

Using HydroSHEDS Free-Flowing Rivers network (Grill et al., 2019), I locate the GRDC gauging stations and find pairs of consecutive stations so that one station is upstream and the other is directly downstream. Then I locate the 144 canal heads from my previous analysis and determine which station pairs have a canal head on the river path in-between them, potentially diverting water and reducing flow at the downstream river station. There are 336 pairs of consecutive gauging stations, out of which 38 have a canal head located in-between.

I employ a triple difference-in-differences identification strategy to estimate the effect of upstream canal presence on river flow downstream. Since canals divert water mostly during periods of peak demand for irrigation water, I compare changes in flows between two consecutive stations, during the peak irrigation months relative to the rest of the year,³⁵ for station pairs with a canal relative to pairs without a canal. The triple-difference strategy is necessary to account for seasonal changes in flow differences between the two stations. If I were to simply compare the average flow in stations downstream of a canal to the average flow in stations upstream, during and outside irrigation season, I would not be able to disentangle the effect of the canal from seasonal factors possibly affecting the magnitude of this difference. For instance, flow in downstream stations might be reduced naturally through evaporation during hot and dry months. Therefore, to difference out the seasonal confounder, I use consecutive gauging stations without canals as a control group for the ones with canals. I estimate the following specification:

$$\Delta \ln Q_{pmy} = \ln Q_{pmy}^{\text{down}} - \ln Q_{pmy}^{\text{up}} = \beta \text{Canal}_p \times \text{Peak}_m + \alpha_p + \lambda_{my} + \epsilon_{pmy} \quad (6)$$

where $\Delta \ln Q_{pmy}$ is the difference in log mean monthly discharges of the downstream and upstream stations in pair p , month m , and year y . Canal_p denotes pairs of stations with a canal head and Peak_m captures the months of peak irrigation demand. Pair fixed effects α_p control for time-invariant characteristics of consecutive station pairs and month-and-year fixed effects λ_{my} control for time trends. Standard errors are clustered two-way at the upstream station and the downstream station level. The coefficient of interest β

³⁴Mean monthly river flow, or mean monthly discharge, measures the average rate at which water flowed past a river gauging station in cubic meters per second (m^3/s).

³⁵The peak irrigation season is defined as July–November, a dry season coupled with wheat growing stage and maize planting stage. Irrigation is most critical when rainfall is insufficient and during the active growth and planting stages of the cropping season.

measures the change in the river flow gap between the downstream stations and upstream stations, during the months of peak irrigation relative to the rest of the year, and for pairs with a canal, relative to pairs without it. The water diversion mechanism predicts $\beta < 0$. The identifying assumption is that, in the absence of canals, the seasonal change (irrigation peak vs. non-peak months) in the gap in flows between downstream and upstream stations should be the same on river segments with canals and river segments without them. To provide evidence in support of this assumption, I plot the month-by-month differences in river flow between downstream and upstream stations, for canal pairs relative to control pairs. Figure 9 shows that these differences for canal pairs and the control pairs evolve in parallel throughout the year, except for the peak irrigation season.

Table 8 reports the estimates. Using all available station pairs for the 1998–2025 period,³⁶ I find that canals reduce flow in downstream stations by 24% (–0.270 log points) during peak irrigation months, relative to the change in flow in control pairs. The effect is even more pronounced in the 2017–2025 period, which was marked by severe droughts and reduced water availability. Canals reduce the flow downstream by 43% (–0.562 log points). The gauging station pairs can be up to 612 km apart.³⁷ As a robustness check, I estimate the effect of canal presence upstream for a restricted sample of pairs of stations that are at most 50 km apart from each other. The effects become slightly larger and remain statistically significant (–0.408 log points and –0.650 log points, a reduction of 34% and 48%).

These three distinct analyses provide evidence in support of the water diversion mechanism. Subsistence farmers in former homelands may be negatively affected by the presence of canals due to canals diverting water away from their traditional source of irrigation water — the rivers.

6 Cost-Benefit Analysis

The high costs and long construction periods associated with large-scale irrigation infrastructure often deter investment. This raises a natural question: do the delayed benefits justify the substantial upfront expenditures? This section addresses this question by comparing the costs of dam construction to the returns generated by such investments. To define benefits, I take advantage of the RD estimates of the effect of dams on annual crop yields. In non-homelands, I found that irrigation dams increase wheat yields by 3.9% on average, and maize yields by 2.3%. This constitutes an intent-to-treat (ITT) effect. To de-

³⁶This is the modern water policy era that started with the enactment of the 1998 National Water Act, which expanded access to water rights.

³⁷The average between-station distance is similar for canal pairs and control pairs.

rive the average treatment effect on the treated (ATT), I scale the ITT by the “first stage”, that is, by the increase of probability in land being irrigated below a canal. I find this first stage effect to be 9.15% (Figure 2). Therefore, the ATT effects are 43% for wheat and 25% for maize. In the former homelands, I found that irrigation dams decrease maize yields by 2.6%. The effect on wheat yields is not statistically significant and so I consider it to be 0%.³⁸ As costs, I use estimates from a feasibility study commissioned by the South African government to evaluate possible solutions to water supply shortages in the Limpopo province (Department of Water Affairs, South Africa, 2014).

The feasibility study evaluates several options and recommends the best one based on the cost of one cubic meter of water supplied — the Paswane Dam. I choose to perform the cost-benefit analysis based on the Paswane Dam and its associated costs since it is likely that the government would choose the option that allows them to supply one cubic meter for the lowest cost. Implementing this infrastructure requires two types of costs: capital costs and operation and maintenance (O&M) costs. Capital costs are further sub-divided into the cost of building the dam, professional fees, social and environmental costs, cost of infrastructure replacement, and cost of land acquisition and relocation. Capital costs are incurred over the duration of the construction, which is ten years in case of the Paswane Dam.³⁹ O&M costs accrue every year since the dam becomes operational (year 10) until the end of the life of the project (45 years later). The detailed costs discounted at 3% are shown in Table 9.

The estimation of benefits relies on several assumptions. First, the RD design identifies only local treatment effects. For the purpose of this cost-benefit analysis I assume that treatment effects extend to all treated areas. Second, I assume there are no general equilibrium effects. For example, changes in crop prices due to an increased production are not allowed. Third, I assume that the Paswane dam is built for irrigation purposes. In practice, a dam can also supply water to industry, but I do not have estimates of treatment effects on non-agricultural sectors. Therefore, net benefits should capture an increase in the value of agricultural production that is causally attributable to an irrigation dam. Finally, I assume that irrigation water from the dam is used to grow annual crops. This is a

³⁸Note that the land cover classification data set does not distinguish between irrigated and non-irrigated subsistence land. I will therefore use the ITT instead of ATT to capture the (dis)benefits that accrue to subsistence farmers.

³⁹I assume that professional fees, social and environmental costs and cost of land acquisition and relocation are paid immediately in the first period, one third of the dam cost is paid in year 3 and the remaining two thirds are paid in year 10, and the infrastructure replacement cost is split in half and paid 15 years and 30 years after the dam becomes operational. I make these assumptions to match the estimates of discounted capital costs from the feasibility study as closely as possible. The authors of the study do not specify how these costs are spread over the construction period.

more conservative scenario since annual crops are relatively low-value compared to other potential uses of irrigation water (vegetables, vines, nut trees). This assumption is also in line with this paper's finding that irrigation dams lead to an expansion of land dedicated to growing annual crops, but not perennial crops.

The Paswane dam would be able to supply 43 million cubic meters per year. I further assume that only 99% of this volume would be used for irrigation, whereas 1% would be for domestic use purposes. This split is the same as for other existing irrigation dams in the area.⁴⁰ Furthermore, from the WARMS database provided by the DWS, I found that the average irrigation water use in the Limpopo province is 14,215 cubic meters per hectare. This implies that the Paswane Dam could irrigate 2,995 hectares of agricultural land. This land would likely be split between commercial and subsistence farmers. I assume that 12% of land is cultivated by subsistence farmers and the remaining 88% by commercial farmers.⁴¹ Finally, I assume that the land used for growing annuals is split between white maize (51%), yellow maize (36%), and wheat (13%). These proportions are based on the national average from the 2017 Census of Commercial Agriculture.

Next, I quantify the additional production in tons for commercial farmers. According to the 2017 Census of Commercial Agriculture, the average yield of rainfed white maize production is 4.54 t/ha. With an ATT of 25%, the additional yield that is attributable to irrigation on existing fields is 1.07 t/ha. Similarly, the additional yields for yellow maize and wheat are 1.26 t/ha and 1.03 t/ha respectively. I also found that irrigation dams lead to an increase in area under production. In such cases, the additional yields attributable to irrigation on new fields is 5.31 t/ha for white maize, 6.26 t/ha for yellow maize, and 3.46 t/ha for wheat.⁴² Note that the expansion of area under production is not captured by the ATT effect on crop yields since I apply a crop mask and, therefore, I only compare productivity on agricultural fields.⁴³ To take into account the fact that irrigation dams lead to an expansion of the area under production, I use my RD estimate of the treatment effect at the extensive margin (increase in the probability of land being cultivated between 1990 and 2020 by 27.8%) and assume that a part of the land cultivated by commercial farmers existed before the dam construction, whereas the other part is newly established fields.⁴⁴

⁴⁰Notably the Albasini Dam and the Tzaneen Dam.

⁴¹This is the average split across the 144 canals in my main analysis sample, that is, within 10 km of distance from each point.

⁴²4.54 t/ha + 1.07 t/ha for white maize. 5 t/ha + 1.26 t/ha for yellow maize. 2.43 t/ha + 1.03 t/ha for wheat.

⁴³If I did not apply the crop mask, then the estimated treatment effect would also capture the fact that areas below canals are more likely to be agricultural fields since the productivity measure would average over cultivated and uncultivated land.

⁴⁴Specifically, I assume that existing fields comprise 2,068 ha and the new fields comprise 554 ha, an increase of 26.8% relative to the existing fields. The remaining 373 ha are assumed to be cultivated by

In my previous analysis, I found that downstream subsistence farmers do not benefit from irrigation dams and additionally experience lower maize yields. This translates into a loss of 0.11 t/ha for white maize and 0.13 t/ha for yellow maize. The RD effect on wheat yields is not statistically significant, therefore, I assume that the effect on wheat production is null. I also found no effect on the extensive margin for subsistence farmers, and thus, all of the land cultivated by subsistence farmers is considered to be existing fields.

Finally, to value the additional production resulting from a new irrigation dam, I collect domestic future prices of maize and wheat from the South Africa Futures Exchange (SAFEX).⁴⁵ I convert the prices to the 2014 South African Rand using the Consumer Price Index and take the average over the past five years (2019–2023).⁴⁶ The additional crop production is then valued at 4,916 R/t for white maize, 4,931 R/t for yellow maize, and 8,021 R/t for wheat. The benefits are discounted using a 3% discount rate and assuming that the production starts after the dam becomes operational (year 10) and lasts for 45 years.

Table 10 reports the benefit-cost ratios for the baseline scenario described above and two counterfactual scenarios. The purpose of these counterfactual scenarios is to assess the sensitivity of the cost benefit analysis to changes in key assumptions.

In the baseline scenario, the project exhibits a benefit-cost ratio of 1.62, implying that large irrigation infrastructure is a cost-effective investment for economic development. The result holds under a more conservative discounting. The benefit-cost ratio in counterfactual 1 with a discount rate of 4% remains well above 1. I also examine the sensitivity to the absence of the extensive margin effect. In counterfactual 2, I assume that the dam benefits only existing agricultural land and no new fields are set up. In this scenario, the benefit-cost ratio falls below 1. Therefore, if the new irrigation infrastructure benefits only existing fields, the investment may not be justified. Policymakers should consider this margin when assessing cost-effectiveness of large irrigation dams.

7 Conclusion

This paper examines distributional consequences of large-scale irrigation infrastructure for commercial and subsistence farmers, exploiting the gravity-driven nature of surface irrigation within a regression discontinuity framework. This comparison is particularly

subsistence farmers.

⁴⁵Historic domestic future prices are made available by the South African Grain Information Service (SAGIS): at https://www.sagis.org.za/safex_historic.html. [Accessed on 25 July 2024.]

⁴⁶The CPI is available at <https://www.statssa.gov.za/publications/P0141/CPIHistory.pdf>. [Accessed on 30 July 2024.]

relevant in the broader context of sub-Saharan Africa, since subsistence farmers in South Africa are representative of the average farmer on the continent. This paper's findings therefore provide insights into the challenges of scaling up this type of infrastructure.

I find a sharp difference in effects of irrigation infrastructure across farmer types. Irrigation canals generate significant productivity gains for commercial farmers in non-homeland areas, raising wheat and maize yields, expanding cultivated area, and reducing unemployment. Subsistence farmers in former homelands, by contrast, experience yield declines and no measurable improvement in labor market outcomes. These distributional impacts reflect the enduring institutional legacy of apartheid: the same infrastructure that creates economic opportunity for well-resourced commercial farmers imposes negative externalities on resource-poor subsistence farmers who lack the capital and institutional support to access canal water and who depend on the river flows that canals divert.

The evidence points to two mechanisms behind the null or negative effects experienced by subsistence farmers. First, capital constraints facing Water User Associations in former homelands prevent the construction and maintenance of the secondary distribution networks needed to connect subsistence farmers to the main canal. Treatment effects in homelands decline monotonically with distance from the canal head, while non-homeland treatment effects are stable across the same gradient. Second, canal intakes divert water from downstream rivers, reducing water availability for subsistence farmers who rely on rivers as their primary irrigation source. Treatment effects in homelands worsen with proximity to rivers. This is confirmed by evidence from hydrological data, showing that canals reduce river flow downstream. Together, these mechanisms imply that the gains from irrigation accrue to farmers with the institutional capacity to access water from canals, while costs are borne disproportionately by farmers without this capacity.

The cost-benefit analysis yields a benefit-cost ratio of 1.62 under a 3 percent discount rate and remains above one at 4 percent, suggesting that large irrigation dams constitute a cost-effective investment when accounting for both commercial gains and subsistence losses. However, this conclusion is sensitive to assumptions about land conversion. In a scenario where new land cannot be converted for agricultural purposes, the ratio falls below one. Two policy implications follow directly from this finding. First, the economic viability of new dam construction depends critically on whether the command area encompasses convertible uncultivated land; investments that cannot expand the agricultural frontier may not recover their costs. Second, subsistence farmers bear negative externalities from large-scale irrigation infrastructure. The 1998 National Water Act explicitly aimed to redress racial inequalities in water access, yet the evidence presented here suggests that structural barriers, such as capital constraints and continued dependence on

river water, have prevented the new policy from successfully improving access to water in the disadvantaged areas.

This paper raises several questions for further research. First, the labor market results show that commercial farm expansion downstream of canals reduces unemployment and the proportion of zero-income households in non-homeland areas, consistent with a local structural transformation in which irrigation-driven agricultural growth generates broader economic activity. Whether this local transformation contributes to economy-wide structural change remains an open question. Second, this paper documents a tension between the focus on smallholder productivity versus promoting agricultural commercialization. This speaks to a broader debate in the development economics literature (Collier and Dercon, 2014; Suri and Udry, 2022). Productivity-enhancing investments that primarily benefit large commercial operations may accelerate structural transformation through a different channel than investments targeting smallholders, but may do so at the cost of worsening inequality in the short run. Disentangling these dynamics, and understanding the conditions under which large-scale irrigation investments can be designed to benefit both commercial and subsistence farmers simultaneously, constitutes an important direction for future research.

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Tables

Table 1: Summary statistics

		RD sample			
	Full sample mean (1)	Control mean (2)	Treatment vs. control diff. (3)	Non-homeland mean (4)	Homeland vs. non- homeland diff. (5)
<i>As share of total area:</i>					
Agri land (share of total area)	0.167	0.215	0.061*** (0.000)	0.233	0.008*** (0.000)
<i>As share of agri land:</i>					
Commercial pivot irrigated	0.158	0.150	0.093*** (0.000)	0.203	-0.201*** (0.001)
Commercial non-pivot irrigated	0.032	0.028	0.029*** (0.000)	0.042	-0.042*** (0.000)
Commercial rainfed	0.51	0.530	-0.044*** (0.000)	0.566	-0.557*** (0.001)
Subsistence	0.089	0.074	-0.019*** (0.000)	0.004	0.681*** (0.000)
Fallow land	0.212	0.218	-0.059*** (0.000)	0.185	0.119*** (0.001)
<i>Geo. Controls</i>					
Terrain Ruggedness Index	2.058	1.202	0.039*** (0.000)	1.184	0.333*** (0.001)
Max. monthly temperature	35.2	36.2	-0.769*** (0.001)	35.9	0.616*** (0.002)
Mean annual precipitation	565	523	29.3*** (0.063)	528	51.9*** (0.101)
Distance to nearest canal	5,834	5,250	-191*** (1.029)	5,097	1,035*** (1.653)
Ditance to nearest river	1,329	1,110	-264*** (0.326)	1,029	-13.4*** (0.532)
Number of obs.	56,465,817	20,008,184	29,049,656	26,428,908	29,049,656
<i>Ag. Productivity</i>					
Wheat season EVI (log)	-1.190	-1.273	0.226*** (0.000)	-1.173	-0.182*** (0.001)
Maize season EVI (log)	-0.539	-0.590	0.140*** (0.000)	-0.516	-0.245*** (0.000)
Number of obs.	9,427,582	4,303,419	6,795,394	6,163,588	6,795,394
<i>Census variables</i>					
% unemployed	0.282	0.292	-0.038*** (0.009)	0.244	0.122*** (0.008)
% hhs with zero income	0.145	0.151	-0.023*** (0.004)	0.139	0.022*** (0.004)
Avg hh income (pc)	33,177	33,625	-1,722 (1,751)	40,957	-25,410*** (1,601)
Avg hh income (pc) - bottom 10%	1,159	1,066	357*** (115)	1,439	-914*** (108)
Number of obs.	4,746	2,247	3,035	2,106	3,035

Notes. This table shows summary statistics for the main outcomes and the control variables for different samples of the data. Column (1) includes all the grid cells within 10km distance of the canals. Columns (2) to (5) include the RD sample of grid cells that are ≤ 50 m of relative elevation to the nearest canal. Column (2) shows the mean of the control group (grid cells above the canals) and column (3) shows the result of the t-test of a difference between treatment and control means (treatment minus control) with standard errors in the parentheses. Column (4) shows the mean of the non-homeland areas and column (5) shows the result of the t-test of a difference between non-homeland and homeland means (homeland minus non-homeland) with standard errors in the parentheses. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 2: Balance in geophysical characteristics

	TRI (1)	Max monthly temp. (°C) (2)	Annual precip. (mm) (3)	Distance to canal (m) (4)	Distance to river (m) (5)	Proportion of sand part. (%) (6)	Proportion of silt part. (%) (7)	Proportion of clay part. (%) (8)
Panel I. All grid cells								
<i>Panel A. Non-homelands</i>								
Below canal	-0.022 (0.023)	0.040 (0.081)	0.406 (0.834)	344*** (94.9)	48.3** (23.3)	0.104 (0.115)	0.010 (0.073)	-0.115 (0.089)
Control mean	1.18	36.1	519	5,149	1,111	59.4	16.7	23.9
R2	0.438	0.692	0.966	0.213	0.191	0.892	0.896	0.835
Clusters	139	139	139	139	139	139	139	139
N	26,425,820	26,425,820	26,425,820	26,425,820	26,425,820	25,442,685	25,442,685	25,442,685
<i>Panel B. Homelands</i>								
Below canal	0.018 (0.067)	0.538* (0.311)	3.43 (4.30)	659*** (226)	195*** (71.1)	-0.467 (0.580)	0.067 (0.159)	0.401 (0.502)
Control mean	1.39	37.0	564	6,196	1,105	56.3	18.0	25.7
R2	0.627	0.744	0.940	0.132	0.260	0.894	0.911	0.783
Clusters	34	34	34	34	34	34	34	34
N	2,619,889	2,619,889	2,619,889	2,619,889	2,619,889	2,416,827	2,416,827	2,416,827
Panel II. Agri. grid cells								
<i>Panel A. Non-homelands</i>								
Below canal	-0.001 (0.028)	0.011 (0.070)	1.93* (1.16)	325** (145)	59.2* (34.6)	-0.040 (0.158)	0.181 (0.115)	-0.142 (0.128)
Control mean	0.885	35.8	549	5,208	1,154	59.6	15.7	24.7
R2	0.455	0.675	0.968	0.320	0.282	0.899	0.911	0.807
Clusters	137	137	137	137	137	137	137	137
N	6,163,457	6,163,457	6,163,457	6,163,457	6,163,457	6,146,023	6,146,023	6,146,023
<i>Panel B. Homelands</i>								
Below canal	-0.142** (0.066)	0.294 (0.210)	2.84 (3.49)	778* (382)	347** (161)	-0.489 (0.376)	-0.141 (0.320)	0.629 (0.462)
Control mean	1.04	37.7	539.3	6,521	1,234	57.9	17.0	25.2
R2	0.604	0.849	0.951	0.142	0.300	0.916	0.929	0.827
Clusters	28	28	28	28	28	28	28	28
N	631,761	631,761	631,761	631,761	631,761	620,484	620,484	620,484

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator obtained by estimating Equation 1 and omitting the control variables. Standard errors are clustered at the canal level. The sample is restricted to grid cells within 10 km of distance and ≤ 50 m of relative elevation to the nearest canal. “TRI” is the terrain ruggedness index, a topographic measure that captures variability of elevation of a given area and is derived from Nunn and Puga (2012). “Max monthly temp. (°C)” is calculated as an average over the maximum temperatures of each month in the period of 2014–2018 and is derived from MODIS Terra Land Surface Temperature dataset. “Annual precip. (mm)” is calculated as an average of total annual rainfall over the period of 2014–2018 and is derived from the CHIRPS dataset (Funk et al., 2015). “Distance to canals (m)” is the distance of the grid cell to the nearest canal. “Distance to river (m)” is the distance of the grid cell to the nearest river (the river network comes from the HydroSHEDS dataset). “Proportion of sand/silt/clay particles (%)” is the percentage of the fine earth fraction made up of sand/silt/clay particles (< 0.002 mm). * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 3: Effects of canals on crop yields and land use

	First stage		Intensive margin		Extensive margin			
	Irrigated annuals (1)	(2)	Wheat EVI (log) (3)	Maize EVI (log) (4)	Annual or fallow (5)	Annual crops (6)	Fallow land (7)	Perennial crops (8)
<i>Panel A. Non-homelands</i>								
Below canal	0.023*** (0.007)	0.018 (0.014)	0.039*** (0.013)	0.023** (0.010)	0.043*** (0.013)	0.045*** (0.013)	-0.003 (0.003)	0.005 (0.004)
Control mean	0.042	0.199	-1.255	-0.562	0.212	0.169	0.044	0.013
R2	0.157	0.476	0.626	0.413	0.161	0.161	0.042	0.255
Clusters	139	137	137	137	139	139	139	139
Crop mask		✓	✓	✓				
N	26,425,820	6,163,457	6,163,451	6,163,457	26,425,820	26,425,820	26,425,820	26,425,820
<i>Panel B. Homelands</i>								
Below canal			-0.029 (0.028)	-0.026** (0.010)	0.011 (0.036)	0.040 (0.030)	-0.033* (0.018)	0.002 (0.006)
Control mean			-1.416	-0.820	0.244	0.161	0.090	0.029
R2			0.778	0.641	0.121	0.119	0.067	0.181
Clusters			28	28	34	34	34	34
Sample			✓	✓				
Crop mask								
N			631,758	631,761	2,619,889	2,619,889	2,619,889	2,619,889

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator obtained by estimating Equation 1 using the SANLC 2018 dataset and the Enhanced Vegetation Index (EVI) derived from the Landsat 8 Collection 1 Tier 1 8-Day EVI composite. Panel A shows the results for non-homelands and Panel B for former homelands. Standard errors are clustered at the canal level. The sample is restricted to grid cells within 10 km of distance and ≤ 50 m of relative elevation to the nearest canal. In columns (2), (3), and (4), the sample includes only grid cells classified as agricultural land in SANLC 2018 (categories 38, 39, 40, 41, 43, 44, 45 which comprise annual crops and corresponding fallow land). In columns (1) and (5)–(8), all grid cells are included. In columns (1) and (2), the dependent variable is a dummy indicating whether the grid cell is classified as growing irrigated annuals (categories 38 and 39). In columns (2) and (3), the dependent variable is EVI which is a remote-sensing measure generated from the Near-Infrared, Red and Blue bands of satellite imagery, and ranges in value from -1 to 1. I extract the maximum EVI over the growing and harvesting seasons of wheat and maize respectively. In columns (5)–(8), the dependent variable is a dummy equal to 1 if land is classified as belonging to a given category in SANLC 2018. “Annual or fallow” is equal to 1 if the grid cell is classified as agricultural (cat. 38, 39, 40, 41, 43, 44, 45 which comprise annual crops and corresponding fallow land), and 0 otherwise. “Annual crops” is equal to 1 if the grid cell is classified growing annual crops (cat. 38, 39, 40, 41 which comprise commercial and subsistence annual crops). “Fallow land” is equal to 1 if the grid cell is classified as fallow land (cat. 43, 44, 45; not including fallow trees). “Perennial crops” is equal to 1 if the grid cell is classified as growing perennial crops (cat. 32, 33, 34, 36, 37 which comprise orchards, sugarcane, and vines) and 0 otherwise. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 4: Effects of canals on land use change between 1990 and 2020

	Expansion between 1990 and 2020 of:			
	Agricultural land (1)	Commercial pivot irr. (2)	Commercial non-pivot (3)	Subsistence (4)
<i>Panel A. Non-homelands</i>				
Below canal	0.010** (0.005)	0.015** (0.006)	0.004** (0.002)	0.000 (0.000)
Control mean	0.035	0.033	0.020	0.000
R2	0.039	0.122	0.014	0.005
Clusters	139	139	139	139
N	26,425,820	26,425,820	26,425,820	26,425,820
<i>Panel B. Homelands</i>				
Below canal	0.006 (0.005)	-0.000 (0.000)	0.002 (0.005)	0.006 (0.005)
Control mean	0.045	0.001	0.026	0.031
R2	0.129	0.018	0.229	0.075
Clusters	34	34	34	34
N	2,619,889	2,619,889	2,619,889	2,619,889

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator obtained by estimating Equation 1 using the SANLC 1990/2020 Change dataset. Panel A shows the results for non-homelands and Panel B for the former homelands. Standard errors are clustered at the canal level. The sample is restricted to grid cells within 10 km of distance and $\leq 50\text{m}$ of relative elevation to the nearest canal. The sample includes both agricultural and non-agricultural grid cells. “Agricultural land” “Any annuals” is equal to 1 if the grid cell is classified as having annual crops in the SANLC 2018 data but not in the SANLC 1990 data, and 0 otherwise. “Commercial pivot irr.” is equal to 1 if the grid cell is classified as commercial pivot-irrigated in the SANLC 2018 data but not in the SANLC 1990 data, and 0 otherwise. “Commercial non-pivot” is equal to 1 if the grid cell is classified as commercial non-pivot irrigated or commercial rainfed in the SANLC 2018 data but not in the SANLC 1990 data and 0 otherwise. “Subsistence” is equal to 1 if the grid cell is classified as subsistence land in the SANLC 2018 data but not in the SANLC 1990 data and 0 otherwise. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 5: Effects of canals on local labor markets

	Unemployed (1)	% HHs with zero inc. (2)	HH inc. pc (3)	HH inc. pc (bottom 10%) (4)
<i>Panel A. Non-homelands</i>				
Below canal	-0.049* (0.028)	-0.027** (0.011)	5980 (10625)	206.5 (547.7)
Control mean	0.250	0.138	45501	1548
R2	0.323	0.191	0.414	0.179
Clusters	110	110	110	110
N	2,103	2,106	2,103	2,103
<i>Panel B. Homelands</i>				
Below canal	-0.020 (0.039)	-0.010 (0.015)	1141 (3568)	398.7 (287)
Control mean	0.377	0.157	16082	553
R2	0.125	0.108	0.306	0.122
Clusters	42	42	42	42
N	927	928	928	928

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator obtained by estimating equation 3. Panel A shows the results for non-homelands and Panel B for the former homelands. Standard errors are clustered at the canal level. The sample is restricted to census small areas that lie within 10 km of distance and $\leq 50\text{m}$ of relative elevation to the nearest canal. “Unemployed” is the small-area-level share of individuals participating in the labor market who are unemployed. “% HHs with zero inc.” is the small-area-level share of households who declare having no income. “HH inc. pc” is the small-area-level average annual household income per capita. “HH inc. pc (bottom 10%)” is the average annual household income per capita, where the small-area-level average is calculated only among the bottom decile of the income distribution. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 6: Heterogeneous treatment effects by distance to canal head and distance to nearest river

	Wheat EVI (log)		Maize EVI (log)	
	(1)	(2)	(3)	(4)
<i>Panel A. Non-homelands</i>				
Below canal	0.040* (0.023)	0.037** (0.018)	0.017 (0.017)	0.026 (0.016)
Below canal x distance to canal head (in km)	-0.000 (0.004)		0.001 (0.003)	
Below canal x distance to nearest river (in km)		0.002 (0.010)		-0.003 (0.013)
Control mean	-1.255	-1.255	-0.562	-0.562
R2	0.625	0.625	0.410	0.410
Clusters	137	137	137	137
N	6,163,451	61,63,451	6,163,457	6,163,457
<i>Panel B. Homelands</i>				
Below canal	0.027 (0.041)	-0.047 (0.030)	0.058* (0.029)	-0.044** (0.017)
Below canal x distance to canal head	-0.009** (0.004)		-0.014** (0.005)	
Below canal x distance to nearest river		0.024* (0.013)		0.025* (0.012)
Control mean	-1.416	-1.416	-0.820	-0.820
R2	0.778	0.778	0.643	0.642
Clusters	28	28	28	28
N	631,758	631,758	631,761	631,761

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator and on the interaction term between the treatment and either distance to the nearest canal or distance to nearest river. Estimates in columns (1) and (3) are obtained by estimating equation 4 and estimates in columns (2) and (4) are obtained by estimating equation 5. The sample is restricted to grid cells classified as agricultural land in SANLC 2018 (categories 38, 39, 40, 41, 43, 44, 45 which comprise annual crops and corresponding fallow land) that lie within 10 km of distance and ≤ 50 m of relative elevation to the nearest canal. Panel A shows the results for non-homelands and Panel B for the former homelands. Standard errors are clustered at the canal level. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 7: Irrigation water sources: homelands vs. non-homelands

Water Resource Type	Homelands	Non-homelands
River	60.8%	38.6%
Borehole	27.6%	32.7%
Surface irrigation scheme	8.6%	25.7%
Other	3.0%	2.9%
N	2,890	52,051

Notes. Own calculations based on *Water Use Authorization and Registration Management System (WARMS), 2024*. The sample is restricted to water users who report irrigated agriculture as their primary use. The category 'other' includes lakes, wetlands, and springs/eyes.

Table 8: Effect of canal presence on river flow downstream

	$\Delta \ln Q_{pym} = \ln Q_{pym}^{\text{down}} - \ln Q_{pym}^{\text{up}}$			
	(1)	(2)	(3)	(4)
Canal between stations $\times$ peak irrigation	-0.270* (0.147)	-0.562*** (0.174)	-0.408* (0.241)	-0.650** (0.254)
Control mean	0.925	0.579	-0.176	-0.814
R2	0.768	0.787	0.782	0.787
Clusters (upstream stations)	232	199	110	96
Clusters (downstream stations)	152	129	94	80
Treated pairs	29	27	8	8
Placebo pairs	211	175	107	91
Restriction on pairs	all	all	≤ 50 km apart	≤ 50 km apart
Period	post-1998	post-2017	post-1998	post-2017
N	48,755	10,085	23,911	4,621

Notes. This table presents triple difference-in-differences estimates of the effect of canal presence upstream on river water flow downstream obtained by estimating equation 6. The coefficient compares the difference in log monthly average flows between two consecutive gauging stations (downstream – upstream), during periods of peak irrigation relative to the rest of the year, for pairs of stations with canals located in-between them relative to pairs without canals. The dependent variable is a difference in log mean monthly discharges, which represent average monthly water flow in a river, measured at monitoring stations in cubic meters per second (m^3/s) and extracted from the Global Runoff Data Center (GRDC). The interaction term Canal between stations $\times$ peak irrigation captures the triple diff-in-diff estimate. The period of peak irrigation is defined as July–November, which corresponds to the dry season combined with the growing stage of wheat and planting stage of maize. Standard errors are clustered two-way at the upstream station level and downstream station level for each pair. The table shows estimates for the period 1999–2025 (columns (1) and (3)), that is, after the enactment of the 1998 National Water Act, and for the period 2018–2025 (columns (2) and (4)), capturing the recent period of droughts. Columns (1) and (2) show estimates for a sample that includes all available pairs of consecutive gauging stations on the same flow path. Columns (3) and (4) restrict the sample to station pairs that are no more than 50 km apart. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table 9: CBA: Breakdown of costs

	Amount (in 2014 Rand)	Discounted amount (in 2014 Rand, total)
<i>Capital costs</i>		
Total	395,419,349	327,221,685
Dam	328,506,564	
Professional fees	39,420,788	
Social & Environmental	3,285,066	
Cost of Infrastructure Replacement	18,416,570	
Cost of Land Acquisition and Relocation	5,790,362	
<i>O&M costs</i>	821,266	15,895,856

Source: Own calculations based on [Department of Water Affairs, South Africa \(2014\)](#). Discount rate is 3%.

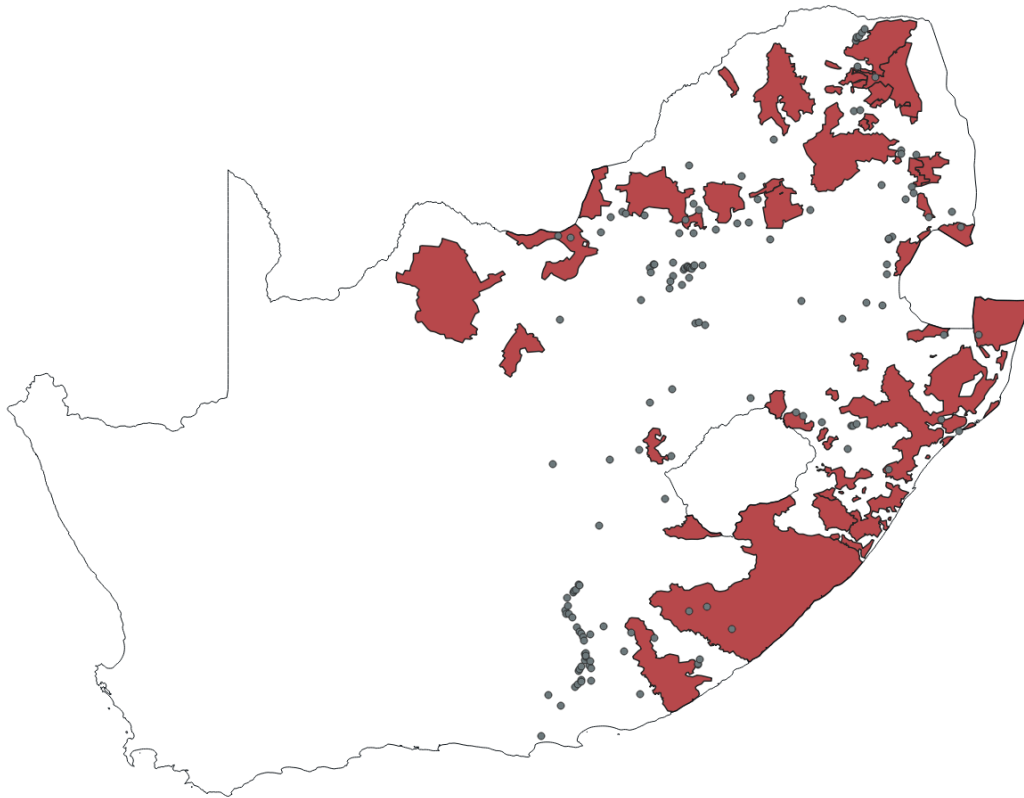
Table 10: CBA: Benefit-cost ratios

Scenario	Discount rate	Discounted benefits (in 2014 Rand)	Discounted costs (in 2014 Rand)	Benefit-cost ratio
Baseline	3%	537,468,348	332,039,758	1.62
Counterfactual 1	4%	420,413,737	321,851,166	1.31
Counterfactual 2	3%	301,065,675	332,039,758	0.91

Source: Own calculations based on [Department of Water Affairs, South Africa \(2014\)](#). Baseline scenario uses RD effects estimated in this paper as benefits. Counterfactual 1 uses 4% discount rate. Counterfactual 2 assumes no extensive margin effect.

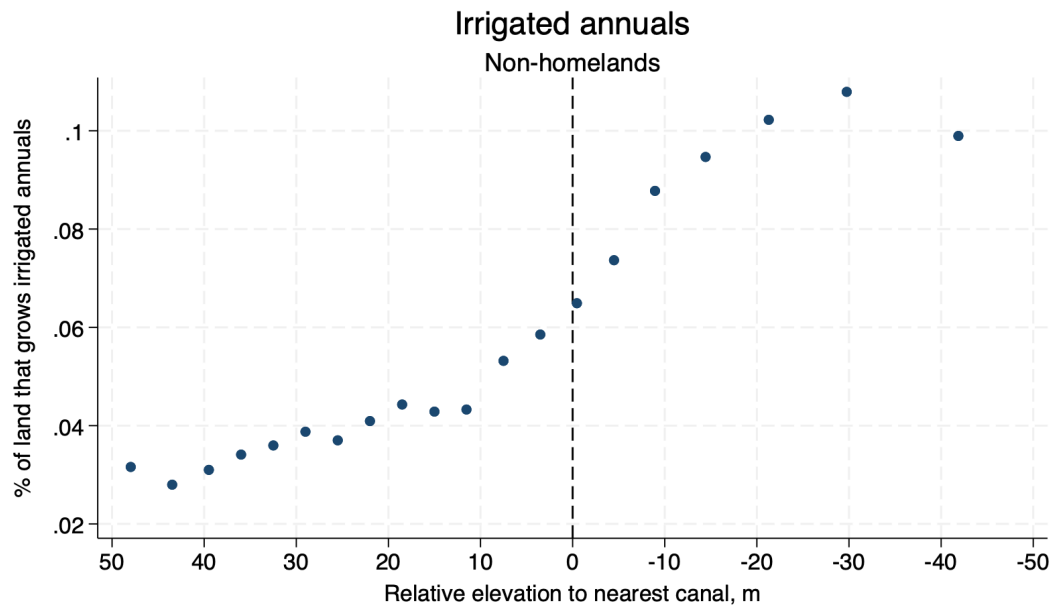
Figures

Figure 1: Spatial distribution of the irrigation canals and the former homelands

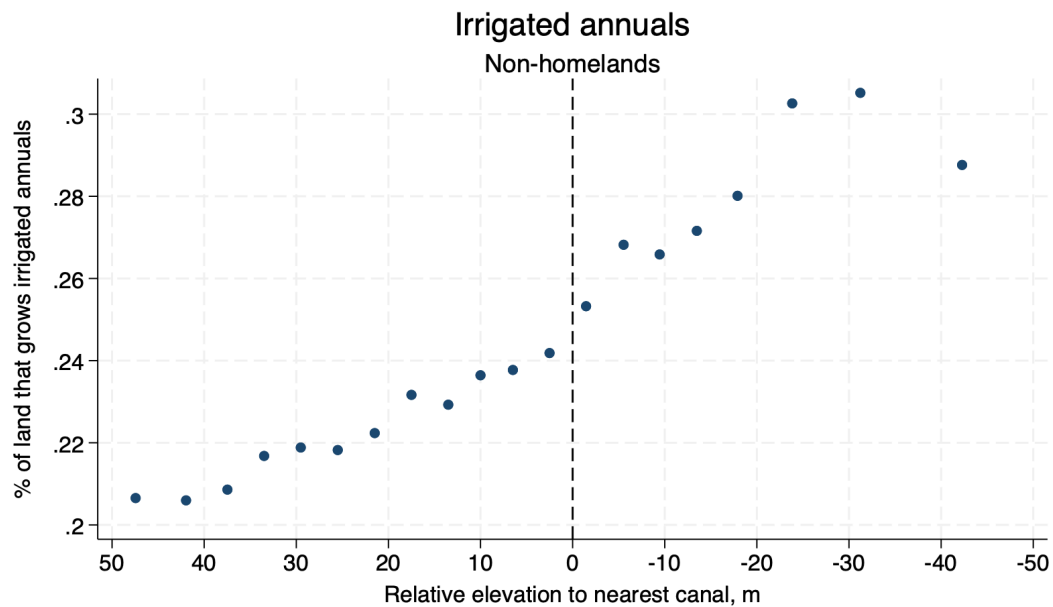


Notes. Map shows the location of the former homelands (in red) and the location of the irrigation canals (grey circles). The canal head coordinates are collected from the South African Department of Water and Sanitation. The boundaries of former homelands come from a shapefile made available by the South African Department of Agriculture, Land Reform, and Rural Development.

Figure 2: First stage: probability of land being irrigated and relative elevation to nearest canal



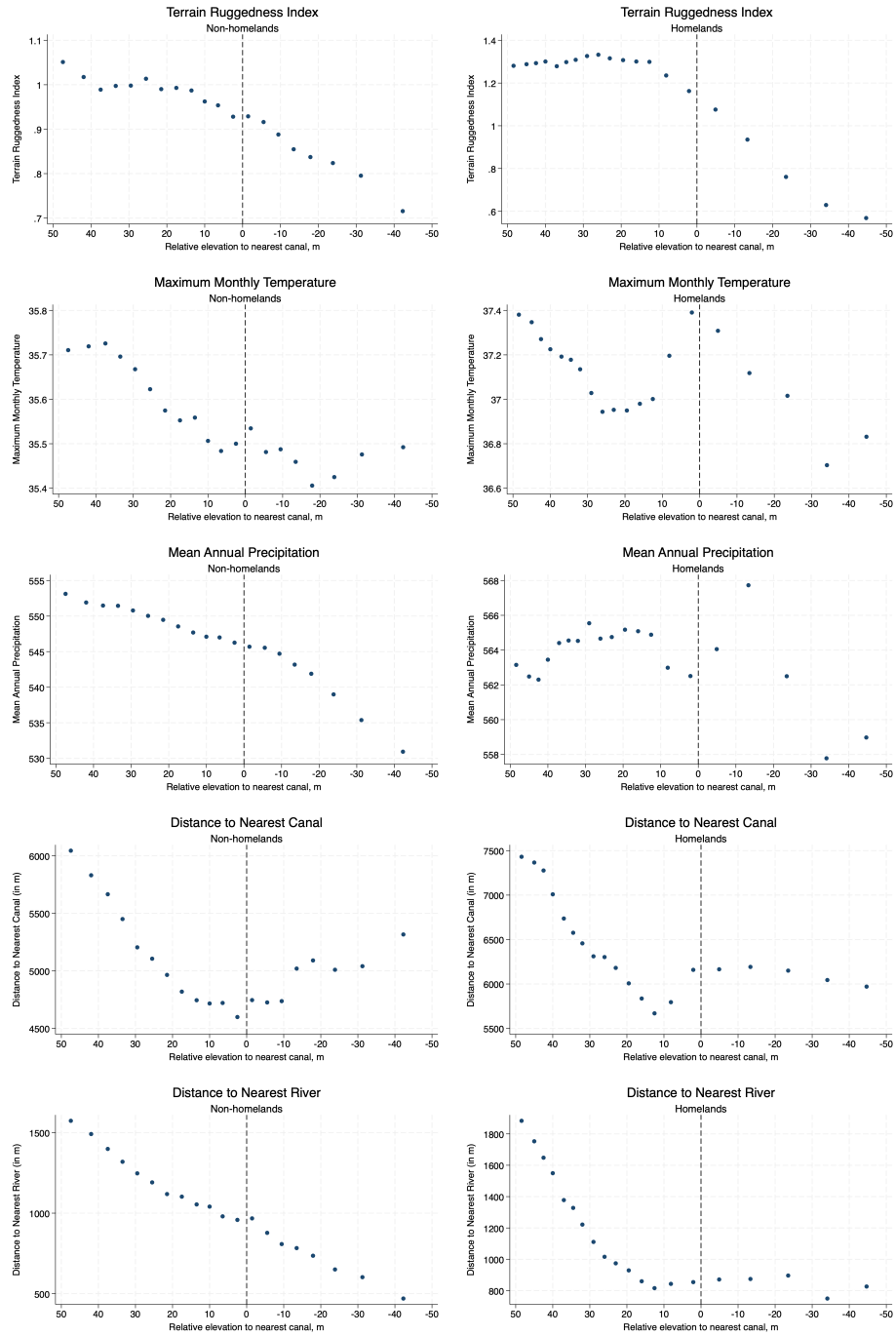
(a) All grid cells



(b) Agricultural grid cells only

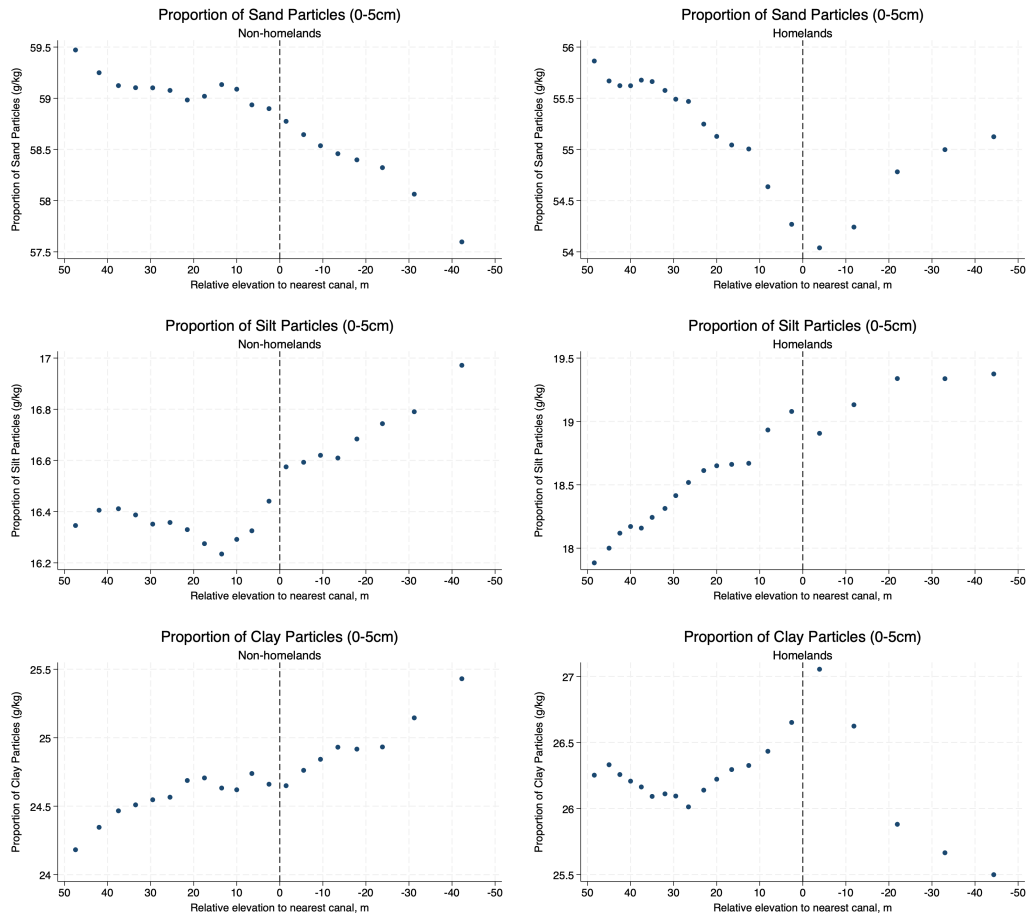
Note. Each panel plots the average outcome (indicator for land growing irrigated annuals) within each of the 50 quantile bins of relative elevation to the nearest canal. The figure includes only non-homelands. The data is residualized on canal fixed effects and five covariates (terrain ruggedness index, distance to nearest canal, distance to nearest river, temperature, and precipitation) using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal).

Figure 3: Continuity through threshold of geophysical characteristics



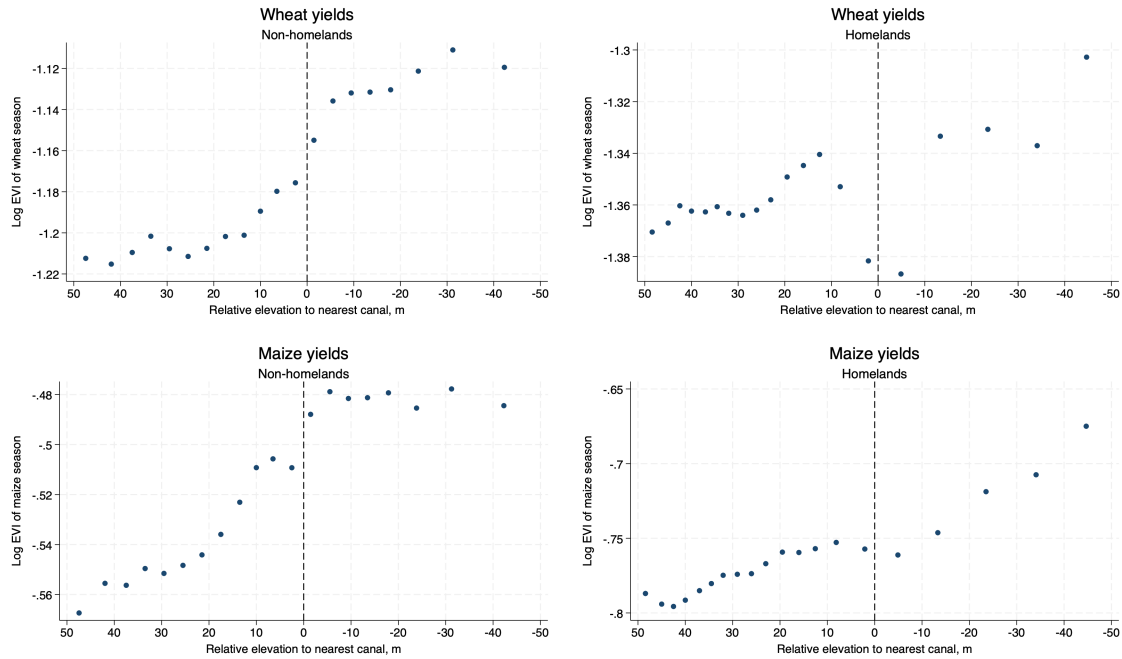
Notes. Each panel plots the average outcome within each of the 50 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). Only agricultural grid cells are included in the analysis.

Figure 4: Continuity through threshold of soil quality variables



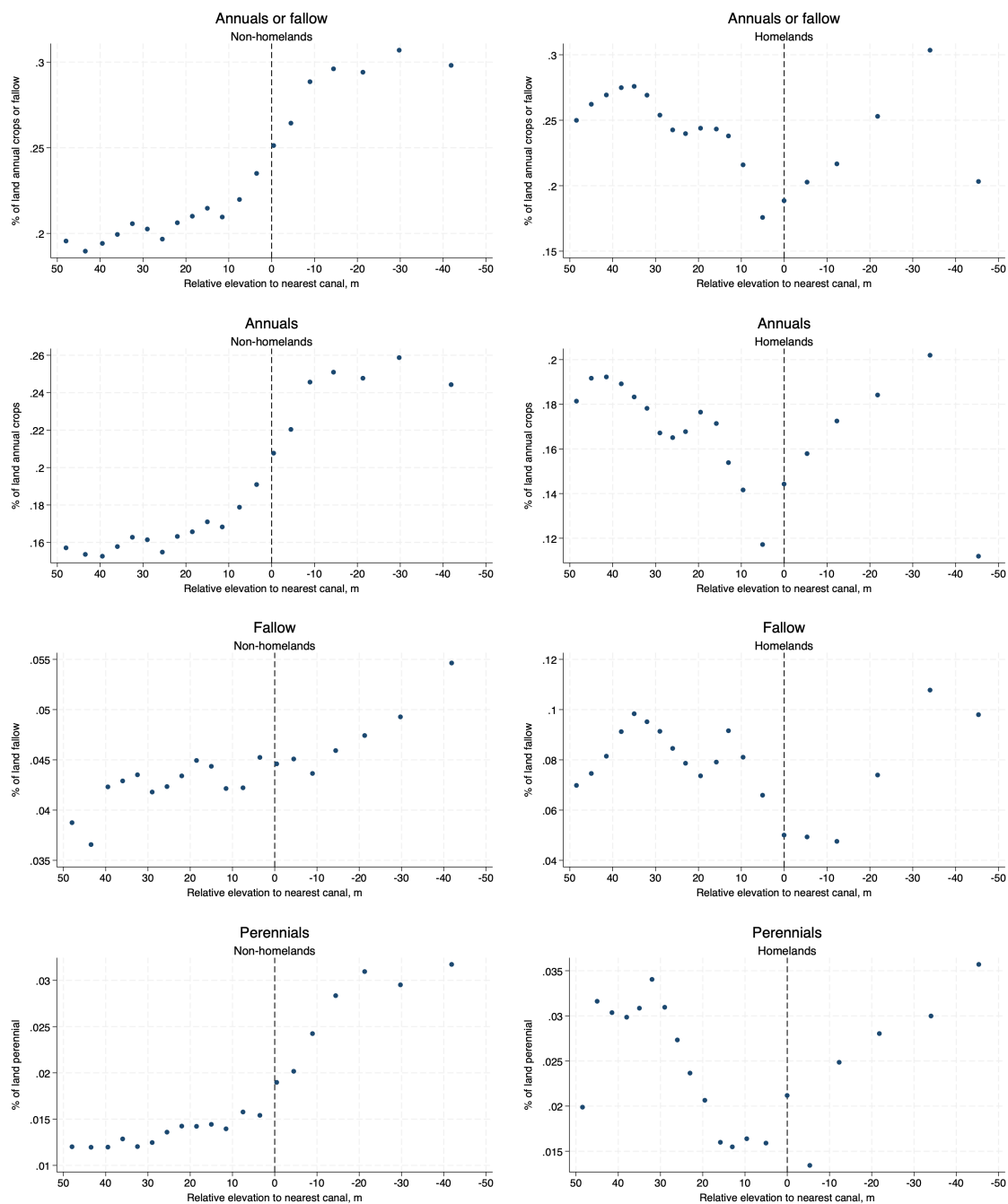
Notes. Each panel plots the average outcome within each of the 20 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). Only agricultural grid cells are included in the analysis.

Figure 5: Regression discontinuity binned scatter plots for agricultural outcomes: intensive margin



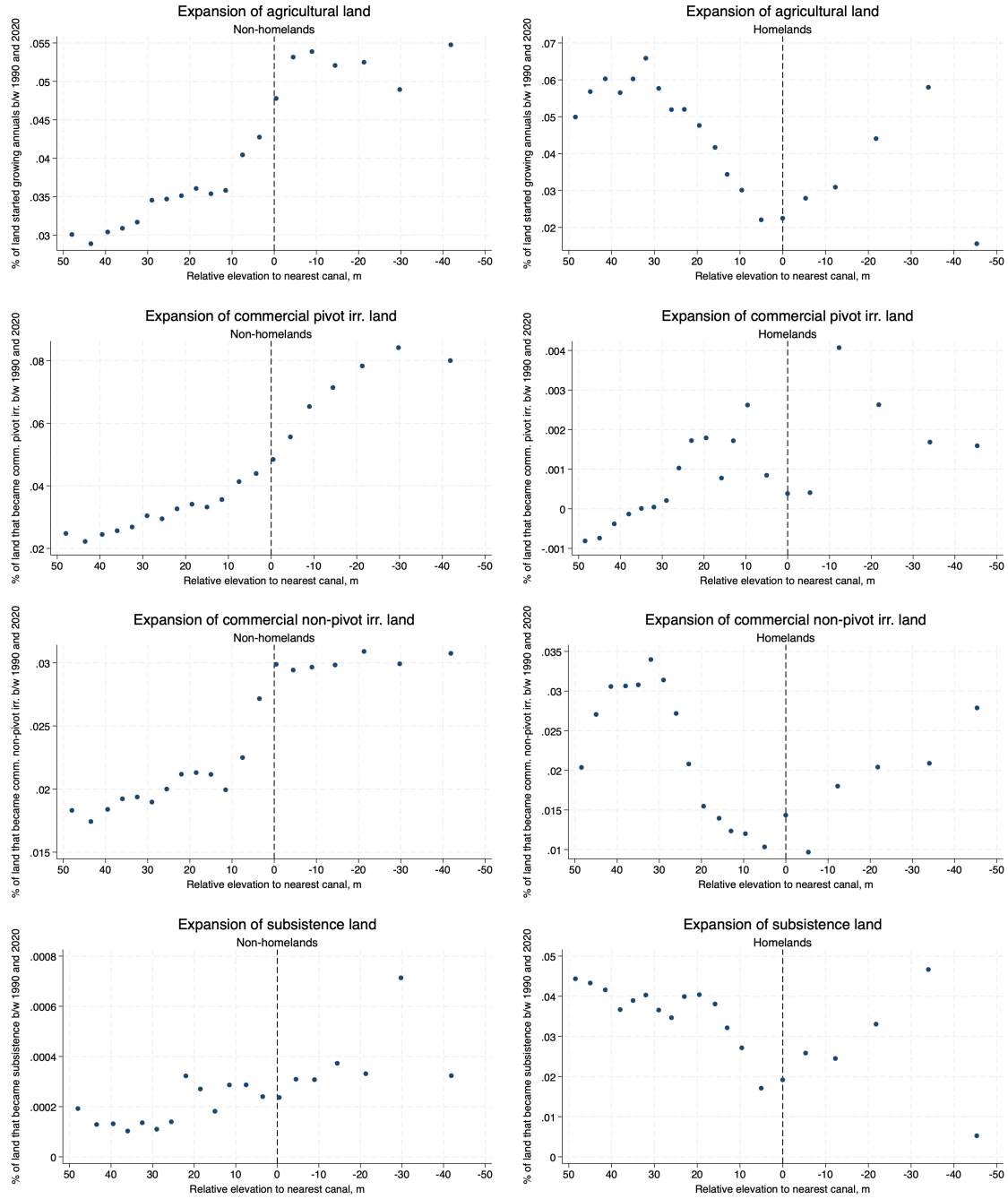
Notes. Each panel plots the average outcome within each of the 20 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects and five covariates (terrain ruggedness index, distance to nearest canal, distance to nearest river, temperature, and precipitation) using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). Crop mask is applied only for wheat and maize yields. The corresponding RD estimates are reported in Table 3.

Figure 6: Regression discontinuity binned scatter plots for agricultural outcomes: extensive margin



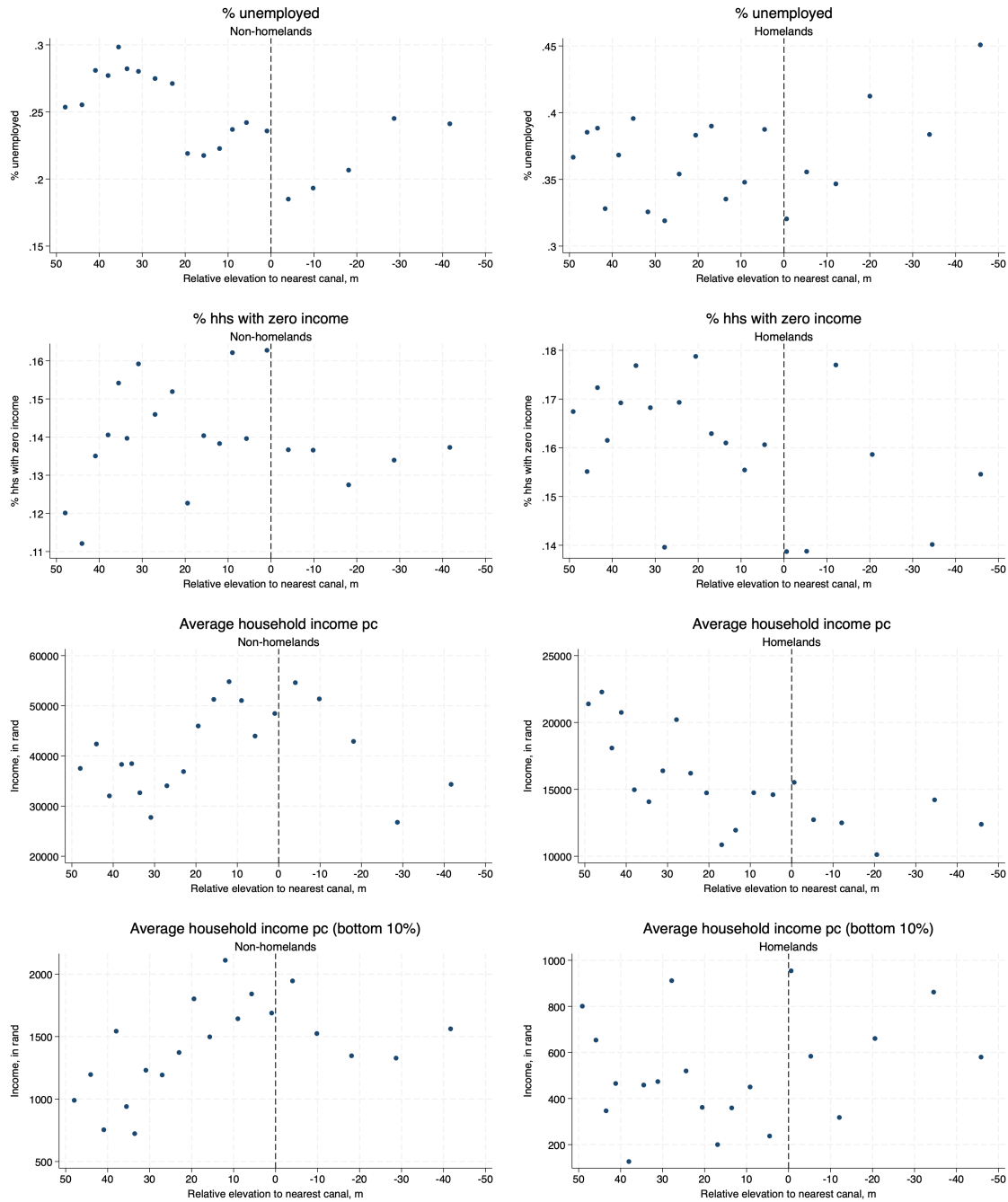
Notes. Each panel plots the average outcome within each of the 20 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects and five covariates (terrain ruggedness index, distance to nearest canal, distance to nearest river, temperature, and precipitation) using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). Crop mask is not applied. The corresponding RD estimates are reported in Table 3.

Figure 7: Regression discontinuity binned scatter plots for land use change 1990/2020



Notes. Each panel plots the average outcome within each of the 20 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects and five covariates (terrain ruggedness index, distance to nearest canal, distance to nearest river, temperature, and precipitation) using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). Crop mask is not applied. The corresponding RD estimates are reported in Table 4.

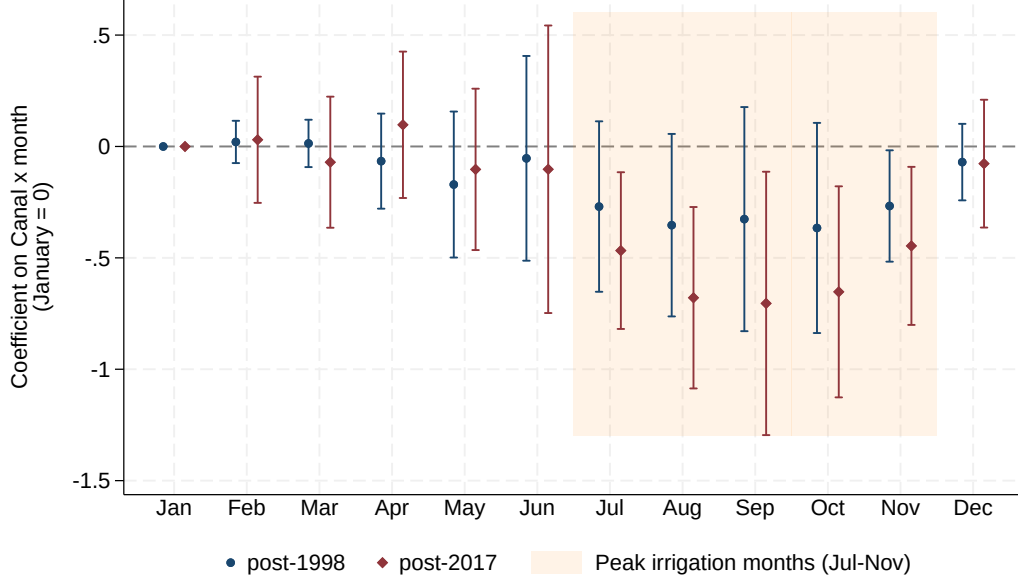
Figure 8: Regression discontinuity binned scatter plots for local labor markets outcomes



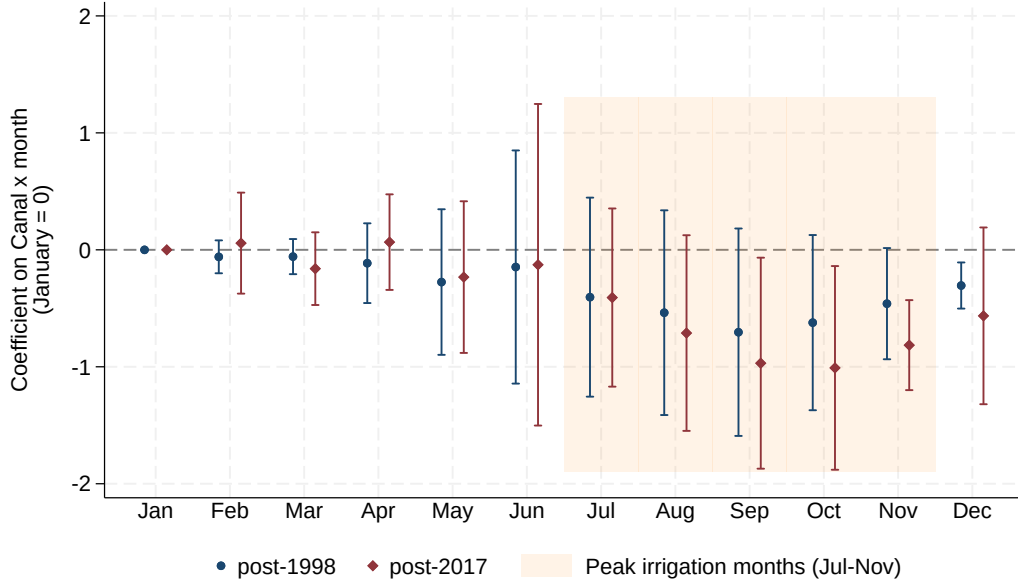
Notes. Each panel plots the average outcome within each of the 20 quantile bins of relative elevation to the nearest canal, separately for non-homelands (left-hand side) and homelands (right-hand side). The data is residualized on canal fixed effects using the `binsreg` package (Cattaneo et al., 2024). The positive values to the left of 0 represent the control units (above a canal) and the negative values to the right of 0 represent the treatment units (below a canal). The corresponding RD estimates are reported in Table 5.

Figure 9: Effect of Canals on River Flow by Calendar Month

(a) All consecutive station pairs



(b) Stations at most 50 km apart



Notes: The figure plots the coefficients β_k from equation $\Delta \ln Q_{pmy} = \sum_{k=2}^{12} \beta_k (\text{Canal}_p \times \mathbf{1}[\text{month} = k]) + \alpha_p + \lambda_{my} + \varepsilon_{pmy}$, where $\mathbf{1}[\text{month} = k]$ is the month indicator and January is the omitted category. The outcome is the difference between log flows of two downstream and upstream consecutive gauging stations. Each coefficient compares a pair of stations with a canal head in-between and pairs without a canal, in calendar month k . The shaded area marks the peak irrigation months (July–November). Bars show 95% confidence intervals, with standard errors two-way clustered by upstream and by downstream station. Blue circles are the estimates from the post-1998 sample, and the red diamonds from the post-2017 sample. Panel (b) keeps only pairs whose stations are at most 50 km apart along the river path.

Appendix

Smallholder Irrigation Schemes

In South Africa, several institutions and policies have been implemented to help smallholder farmers benefit from irrigation. One key initiative has been the establishment of smallholder irrigation schemes, which aim to improve rural livelihoods by providing access to water for agriculture. However, some agronomists argue that the development goals of these schemes have largely remained unfulfilled due to poor farming practices, such as inadequate weed and pest control and failure to increase cropping intensity (Fanadzo, Chiduza and Mnkeni, 2018).

Currently, there are around 330 smallholder irrigation schemes located mostly in the former homelands, covering 50,000 hectares and supporting an estimated 200,000 to 250,000 smallholder farmers. The history of smallholder irrigation development in South Africa can be divided into four distinct periods. First, during the 19th century, the early smallholder irrigation development consisted in colonists transferring irrigation technology to locals, but most such projects had ceased to exist by the end of the century. During the smallholder canal scheme era (1930–1960), the focus was on providing black families in the homelands with livelihoods based on farming. Most schemes constructed concrete weirs to divert water from rivers, and some large dams were also built. Next, during the homeland era (1960–1990), the South African government funded large irrigation schemes with the goal of economic development in the homelands. Over time, this infrastructure proved too costly to maintain and fell into disrepair. Finally, the irrigation management transfer and revitalization era started in the early 1990s with the end of the apartheid regime. Government involvement has decreased, leading to a shift towards local management of irrigation through catchment management agencies, creation of water users associations, and transfers of water rights. The main focus is on repairing the existing infrastructure, devising better maintenance strategies, and training farmers on best cultivation practices.

Table A–1: Robustness of the intensive margin results: extending to multiple years

	Intensive margin (2000-2019)	
	Wheat EVI (log) (1)	Maize EVI (log) (2)
<i>Panel A. Non-homelands</i>		
Below canal	0.022* (0.013)	0.009 (0.006)
Control mean	-1.495	-0.778
R2	0.578	0.331
Canal FE	Yes	Yes
Year FE	Yes	Yes
Clusters	137	137
N	63,065,671	63,410,102
<i>Panel B. Homelands</i>		
Below canal	-0.058* (0.034)	-0.036*** (0.010)
Control mean	-1.489	-0.894
R2	0.615	0.451
Canal FE	✓	✓
Year FE	✓	✓
Clusters	27	27
N	1,466,412	1,464,739

Notes. This table reports regression discontinuity estimates of the coefficients on the treatment indicator obtained by estimating Equation 1 using the Enhanced Vegetation Index (EVI) derived from the Landsat 8 Collection 1 Tier 1 8-Day EVI composite and the UMD GLAD Global Cropland Map (Potapov et al., 2022). Regressions include year fixed effects. The sample includes only grid cells classified as cropland by the UMD GLAD Global Cropland Map (annuals, perennials, forage, biofuel). Enhanced Vegetation Index (EVI) is a remote-sensing measure that is generated from the Near-IR, Red and Blue bands of each satellite image, and ranges in value from -1 to 1. It is derived from the Landsat 8 Collection 1 Tier 1 8-Day EVI composite. I extract the maximum EVI over the growing and harvesting seasons of wheat and maize respectively. * significant at 10%, ** significant at 5%, *** significant at 1%.

Table A–2: Comparison of UMD GLAD 2019 and SANLC 2018

	UMD GLAD 2019 cropland by NLC category
<i>Panel A. Non-homelands</i>	
Commercial Annuals Crops Rain-Fed / Dryland / Non-Irrigated	49%
Commercial Annuals Pivot Irrigated	32%
Commercial Annuals Non-Pivot Irrigated	6%
Natural Grassland	4%
Cultivated Commercial Sugarcane Non-Pivot	2%
<i>Panel B. Homelands</i>	
Subsistence / Small-Scale Annual Crops	32%
Cultivated Emerging Farmer Sugarcane Non-Pivot	25%
Open Woodland	12%
Cultivated Commercial Sugarcane Non-Pivot	4%
Residential Formal (Bush)	4%

Notes. The table compares the grid cells defined as cropland in the UMD GLAD 2019 global cropland map with the SANLC 2018 land cover classes. Panel A represents cropland within non-homeland areas, while Panel B shows cropland within homeland areas. Only the five most frequent land cover categories within each area are reported. Percentages indicate the proportion of UMD GLAD 2019 cropland grid cells that fall into each SANLC 2018 land cover category within the respective regions. Only five most frequent categories for both homelands and non-homelands are included.

Table A-3: Robustness of the intensive margin results: different vertical bandwidths

	Wheat EVI (log)				Maize EVI (log)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.021 (0.013)	0.039*** (0.012)	0.044*** (0.013)	0.052*** (0.013)	0.010 (0.009)	0.023** (0.010)	0.032*** (0.011)	0.035*** (0.012)
Control mean	-1.204	-1.255	-1.272	-1.271	-0.525	-0.562	-0.576	-0.579
R2	0.621	0.625	0.625	0.623	0.391	0.410	0.420	0.421
Clusters	137	137	137	138	137	137	137	138
Vertical bandwidth	25	50	75	100	25	50	75	100
N	3,792,405	6,163,451	7,342,030	7,848,264	3,792,407	6,163,457	7,342,038	7,848,272
<i>Panel B. Homelands</i>								
Below canal	-0.037 (0.022)	-0.029 (0.028)	-0.067** (0.030)	-0.057** (0.024)	-0.018** (0.006)	-0.026** (0.010)	-0.025* (0.012)	-0.008 (0.015)
Control mean	-1.397	-1.416	-1.414	-1.399	-0.794	-0.820	-0.817	-0.805
R2	0.765	0.778	0.777	0.770	0.658	0.641	0.637	0.634
Clusters	26	28	31	32	26	28	31	32
Vertical bandwidth	25	50	75	100	25	50	75	100
N	279,188	631,758	831,456	908,112	279,188	631,761	831,459	908,115

Notes. See notes for Table 3. The robustness of the intensive margin results is shown for vertical bandwidths of 25 m, 75 m, and 100 m. The preferred specification vertical bandwidth (50 m) is shown for comparison.

Table A-4: Robustness of the extensive margin results: different vertical bandwidths

	Agricultural land				Annual crops			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.020* (0.011)	0.043*** (0.013)	0.053*** (0.013)	0.053*** (0.014)	0.020* (0.011)	0.045*** (0.013)	0.054*** (0.014)	0.053*** (0.014)
Control mean	0.227	0.212	0.200	0.189	0.184	0.169	0.158	0.149
R2	0.162	0.161	0.166	0.170	0.164	0.161	0.160	0.163
Clusters	138	139	139	139	138	139	139	139
Vertical bandwidth	25	50	75	100	25	50	75	100
N	15,201,625	26,425,820	33,427,200	37,590,184	15,201,625	26,425,820	33,427,200	37,590,184
<i>Panel B. Homelands</i>								
Below canal	0.001 (0.021)	0.011 (0.036)	-0.008 (0.036)	-0.034 (0.035)	0.025 (0.021)	0.040 (0.030)	0.017 (0.028)	-0.003 (0.027)
Control mean	0.212	0.244	0.242	0.228	0.139	0.161	0.161	0.152
R2	0.115	0.121	0.135	0.142	0.120	0.119	0.127	0.133
Clusters	33,000	34,000	36,000	37,000	33,000	34,000	36,000	37,000
Vertical bandwidth	25	50	75	100	25	50	75	100
N	1,313,787	2,619,889	3,494,244	3,961,235	1,313,787	2,619,889	3,494,244	3,961,235

Notes. See notes for Table 3. The robustness of the extensive margin results is shown for vertical bandwidths of 25 m, 75 m, and 100 m. The preferred specification vertical bandwidth (50 m) is shown for comparison.

Table A-5: Robustness of the intensive margin results: different horizontal bandwidths

	Wheat EVI (log)				Maize EVI (log)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.042** (0.017)	0.036** (0.018)	0.041*** (0.014)	0.039*** (0.013)	0.002 (0.025)	0.017 (0.016)	0.024** (0.012)	0.023** (0.010)
Control mean	-1.082	-1.158	-1.221	-1.255	-0.506	-0.518	-0.543	-0.562
R2	0.714	0.652	0.632	0.625	0.501	0.444	0.414	0.410
Clusters	130	132	137	137	130	132	137	137
Horizontal bandwidth	2500	5000	7500	10000	2500	5000	7500	10000
N	1,256,639	3,088,758	4,784,964	6,163,451	1,256,639	3,088,761	4,784,969	6,163,457
<i>Panel B. Homelands</i>								
Below canal	-0.047 (0.044)	-0.021 (0.033)	-0.013 (0.036)	-0.029 (0.028)	-0.064 (0.038)	-0.013 (0.027)	-0.014 (0.013)	-0.026** (0.010)
Control mean	-1.516	-1.510	-1.443	-1.416	-0.889	-0.885	-0.830	-0.820
R2	0.826	0.775	0.750	0.778	0.716	0.670	0.647	0.641
Clusters	17	20	27	28	17	20	27	28
Horizontal bandwidth	2500	5000	7500	10000	2500	5000	7500	10000
N	40,292	181,745	395,794	631,758	40,292	181,748	395,797	631,761

Notes. See notes for Table 3. The robustness of the intensive margin results is shown for horizontal bandwidths (the distance to the nearest canal) of 2500 m, 5000 m, and 7500 m. The preferred specification horizontal bandwidth (10000 m) is shown for comparison.

Table A-6: Robustness of the extensive margin results: different horizontal bandwidths

	Agricultural land			Annual crops				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.023 (0.020)	0.037** (0.015)	0.048*** (0.014)	0.043*** (0.013)	0.023 (0.020)	0.037** (0.015)	0.051*** (0.014)	0.045*** (0.013)
Control mean	0.221	0.203	0.209	0.212	0.183	0.167	0.168	0.169
R2	0.157	0.156	0.161	0.161	0.161	0.159	0.161	0.161
Clusters	134	134	138	139	134	134	138	139
Horizontal bandwidth	2500	5000	7500	10000	2500	5000	7500	10000
N	5,019,861	13,351,772	20,579,592	26,425,820	5,019,861	13,351,772	20,579,592	26,425,820
<i>Panel B. Homelands</i>								
Below canal	-0.017 (0.048)	0.016 (0.039)	0.019 (0.035)	0.011 (0.036)	-0.024 (0.040)	0.030 (0.035)	0.045 (0.029)	0.040 (0.030)
Control mean	0.164	0.213	0.225	0.244	0.132	0.156	0.151	0.161
R2	0.198	0.145	0.119	0.121	0.195	0.139	0.121	0.119
Clusters	21	26	33	34	21	26	33	34
Horizontal bandwidth	2500	5000	7500	10000	2500	5000	7500	10000
N	238,859	848,240	1,717,153	2,619,889	238,859	848,240	1,717,153	2,619,889

Notes. See notes for Table 3. The robustness of the extensive margin results is shown for horizontal bandwidths (the distance to the nearest canal) of 2500 m, 5000 m, and 7500 m. The preferred specification horizontal bandwidth (10000 m) is shown for comparison.

Table A-7: Robustness of the intensive margin results: donut hole

	Wheat EVI (log)			Maize EVI (log)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.047*** (0.016)	0.055*** (0.018)	0.061*** (0.020)	0.066*** (0.022)	0.026** (0.012)	0.027* (0.014)	0.029* (0.016)	0.030 (0.018)
Control mean	-1.262	-1.267	-1.272	-1.278	-0.568	-0.571	-0.576	-0.581
R2	0.627	0.630	0.631	0.633	0.414	0.417	0.421	0.425
Clusters	137	137	137	137	137	137	137	137
Donut size	2	4	6	8	2	4	6	8
N	5,754,599	5,434,303	5,108,714	4,783,933	5,754,605	5,434,307	5,108,718	4,783,937
<i>Panel B. Homelands</i>								
Below canal	-0.028 (0.032)	-0.026 (0.033)	-0.020 (0.034)	-0.013 (0.034)	-0.030** (0.012)	-0.032** (0.013)	-0.032* (0.016)	-0.034* (0.019)
Control mean	-1.417	-1.417	-1.418	-1.419	-0.822	-0.824	-0.825	-0.827
R2	0.779	0.779	0.780	0.781	0.642	0.643	0.643	0.643
Clusters	28	28	28	28	28	28	28	28
Donut size	2	4	6	8	2	4	6	8
N	608,750	590,832	571,526	551,885	608,753	590,835	571,529	551,888

Notes. See notes for Table 3. The robustness of the intensive margin results is shown for donut holes (in the relative elevation to the nearest canal) of 2 m, 4 m, 6 m, and 8 m.

Table A-8: Robustness of the extensive margin results: donut hole

	Agricultural land				Annual crops			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.054*** (0.015)	0.065*** (0.017)	0.073*** (0.020)	0.075*** (0.022)	0.057*** (0.015)	0.067*** (0.017)	0.076*** (0.019)	0.079*** (0.021)
Control mean								
R2	0.209	0.208	0.206	0.205	0.166	0.165	0.163	0.162
Clusters	0.162	0.163	0.165	0.167	0.161	0.163	0.164	0.166
Donut size	139	139	139	139	139	139	139	139
N	2	4	6	8	2	4	6	8
	24,842,043	23,590,233	22,331,548	21,091,141	24,842,043	23,590,233	22,331,548	21,091,141
<i>Panel B. Homelands</i>								
Below canal	0.016 (0.043)	0.017 (0.048)	0.019 (0.055)	0.027 (0.062)	0.050 (0.035)	0.056 (0.039)	0.060 (0.044)	0.069 (0.051)
Control mean								
R2	0.247	0.249	0.252	0.253	0.163	0.164	0.166	0.167
Clusters	0.119	0.119	0.119	0.121	0.119	0.120	0.121	0.122
Donut size	34	34	34	34	34	34	34	34
N	2	4	6	8	2	4	6	8
	2,494,570	2,390,651	2,287,231	2,191,794	2,494,570	2,390,651	2,287,231	2,191,794

Notes. See notes for Table 3. The robustness of the extensive margin results is shown for donut holes (in the relative elevation to the nearest canal) of 2 m, 4 m, 6 m, and 8 m.

Table A-9: Robustness of the intensive margin results: control variables and fixed effects

	Wheat EVI (log)			Maize EVI (log)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Non-homelands								
Below canal	0.036** (0.014)	0.037*** (0.012)	0.037*** (0.012)	0.036*** (0.013)	0.025** (0.010)	0.022** (0.010)	0.022** (0.010)	0.021* (0.011)
Control mean	-1.255	-1.255	-1.255	-1.255	-0.562	-0.562	-0.562	-0.562
R2	0.604	0.626	0.632	0.611	0.389	0.413	0.429	0.395
Clusters	137	137	137	137	137	137	137	137
Weather & terrain controls		✓	✓	✓		✓	✓	✓
Soil quality controls		✓	✓			✓	✓	
Control for carbon & nitrogen			✓				✓	
Canal FE		✓	✓			✓		
District FE				✓				✓
N	6,163,582	6,146,017	6,139,658	6,163,451	6,163,588	6,146,023	6,139,664	6,163,457
Panel B. Homelands								
Below canal	-0.023 (0.034)	-0.031 (0.026)	-0.035 (0.026)	-0.029 (0.028)	-0.020* (0.012)	-0.027** (0.011)	-0.031*** (0.011)	-0.029*** (0.009)
Control mean	-1.416	-1.416	-1.416	-1.416	-0.820	-0.820	-0.820	-0.820
R2	0.768	0.784	0.786	0.776	0.636	0.648	0.650	0.636
Clusters	28	28	28	28	28	28	28	28
Weather & terrain controls		✓	✓	✓		✓	✓	✓
Soil quality controls		✓	✓			✓	✓	
Control for carbon & nitrogen			✓				✓	
Canal FE		✓	✓			✓		
Munic. FE				✓				✓
N	631,803	620,484	618,006	631,758	631,806	620,484	618,006	631,761

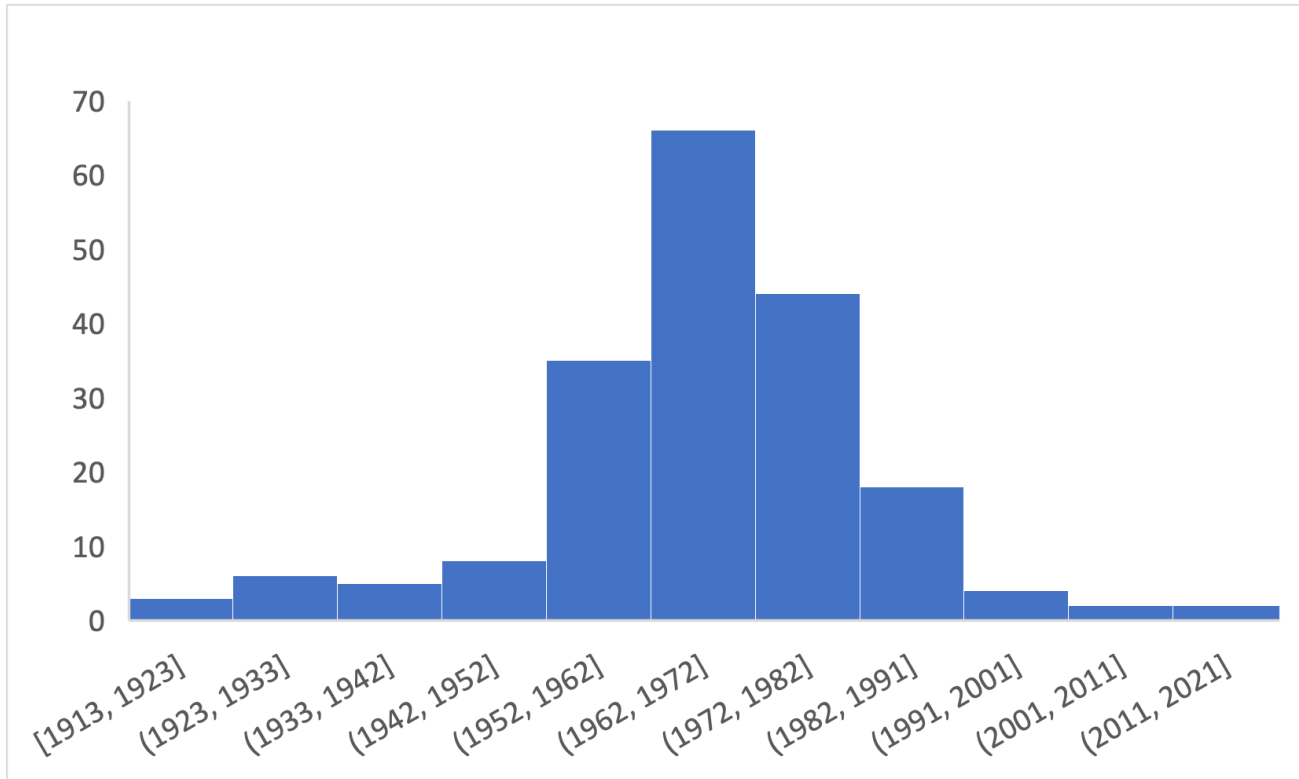
Notes. See notes for Table 3. The robustness of the intensive margin results is shown for different sets of control variables and different level of fixed effects. Columns (1) and (5) show results of RD regressions without including the covariates from the preferred specification (weather & terrain controls). Columns (2) and (6) show results of RD regressions with the covariates from the preferred specification and soil quality controls. Columns (3) and (7) show results of RD regressions with the covariates from the preferred specification, soil quality controls, and carbon and nitrogen controls. Columns (4) and (8) show the results of RD regressions with the covariates from the preferred specification and local municipality FE replacing the canal FE.

Table A–10: Robustness of the extensive margin results: control variables and fixed effects

	Agricultural land			Annual crops				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Non-homelands</i>								
Below canal	0.046*** (0.013)	0.034*** (0.012)	0.029** (0.012)	0.045*** (0.014)	0.048*** (0.013)	0.036*** (0.012)	0.032*** (0.011)	0.048*** (0.013)
Control mean	0.212	0.212	0.212	0.212	0.169	0.169	0.169	0.169
R2	0.147	0.175	0.194	0.133	0.144	0.175	0.197	0.132
Clusters	139,000	139,000	139,000	139,000	139,000	139,000	139,000	139,000
Weather & terrain controls		✓	✓	✓		✓	✓	✓
Soil quality controls		✓	✓			✓	✓	
Control for carbon & nitrogen			✓				✓	
Canal FE	✓	✓	✓		✓	✓	✓	
Munic. FE				✓				✓
N	26,428,908	25,442,685	25,346,761	26,420,063	26,428,908	25,442,685	25,346,761	26,420,063
<i>Panel B. Homelands</i>								
Below canal	0.036 (0.039)	-0.028 (0.031)	-0.027 (0.031)	0.008 (0.035)	0.053 (0.033)	0.009 (0.027)	0.010 (0.027)	0.038 (0.029)
Control mean	0.244	0.244	0.244	0.244	0.161	0.161	0.161	0.161
R2	0.096	0.129	0.130	0.112	0.103	0.129	0.129	0.112
Clusters	34,000	34,000	34,000	34,000	34,000	34,000	34,000	34,000
Weather & terrain controls		✓	✓	✓		✓	✓	✓
Soil quality controls		✓	✓			✓	✓	
Control for carbon & nitrogen			✓				✓	
Canal FE	✓	✓	✓		✓	✓	✓	
Munic. FE				✓				✓
N	2,620,748	2,416,827	2,398,606	2,619,578	2,620,748	2,416,827	2,398,606	2,619,578

Notes. See notes for Table 3. The robustness of the extensive margin results is shown for different sets of control variables and different level of fixed effects. Columns (1) and (5) show results of RD regressions without including the covariates from the preferred specification (weather & terrain controls). Columns (2) and (6) show results of RD regressions with the covariates from the preferred specification and soil quality controls. Columns (3) and (7) show results of RD regressions with the covariates from the preferred specification, soil quality controls, and carbon and nitrogen controls. Columns (4) and (8) show the results of RD regressions with the covariates from the preferred specification and local municipality FE replacing the canal FE.

Figure A–1: Histogram of years of construction of large irrigation dams



Notes. The data comes from AQUASTAT, the geo-referenced database on dams in Africa. Only dams with the listed purpose of irrigation are included. There are 200 irrigation dams and 550 dams in total.

Figure A–2: Schematic representation of elevation-based RDD

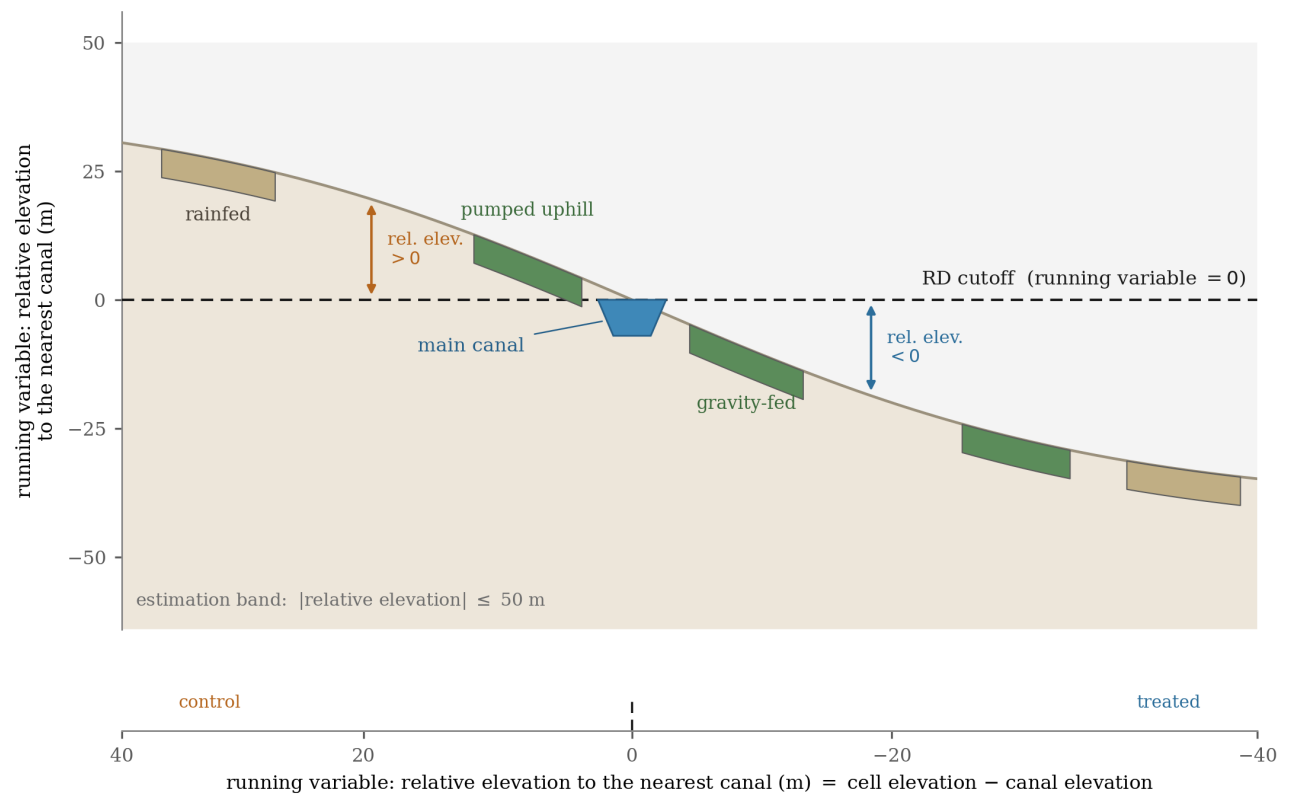


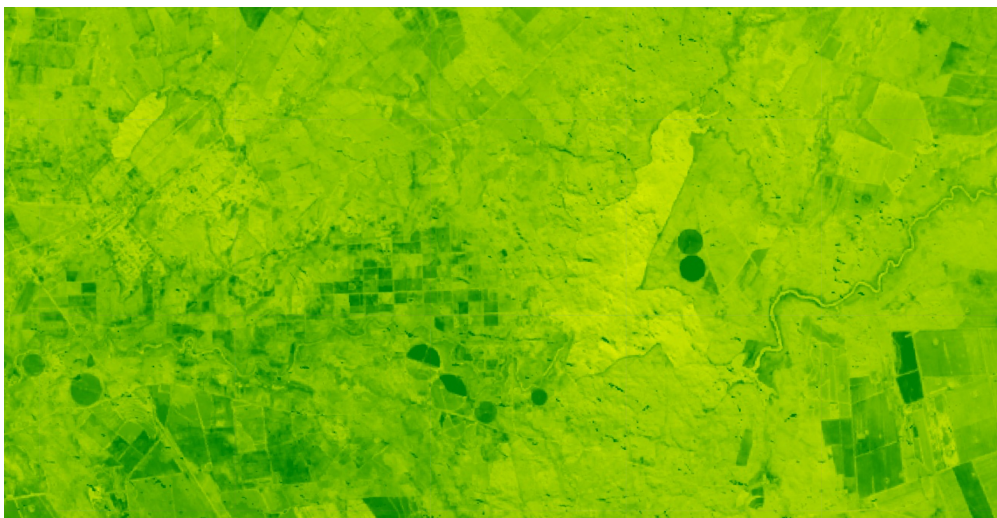
Figure A-3: Visualization of the main outcome variables



(a) Raw satellite image



(b) SANLC 2018



(c) Landsat 8 EVI composite